A Project report on

REAL-TIME ACCIDENT MONITORING SYSTEM USING IOT

Submitted in partial fulfillment of the requirements for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION ENGINEERING

by

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Rotarypuram Village, B K Samudram Mandal, Ananthapuramu - 515701

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DECLARATION CERTIFICATE

We students of **Electronics and Communication Engineering, SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY (AUTONOMOUS),** Rotarypuram, hereby declare that the dissertation entitled **"REAL-TIME ACCIDENT MONITORING SYSTEM USING IOT"** embodies the report of our project work carried out by us during IV year under the guidance of Dr. K. Dadasikandar, M. Tech, Ph. D., Associate Professor, Department of Electronics and Communication Engineering, Srinivasa Ramanujan Institute of Technology, and this work has been submitted for the partial fulfillment of the requirements for the award of degree of Bachelor of Technology.

The results embodied in this project report have not been submitted to any other University or Institute for the award of any Degree or Diploma.

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Project Associates

ABSTRACT

Road accidents are a major global concern, often resulting in severe injuries, fatalities, and economic losses. The real-time accident monitoring system using the Internet of Things (IoT) is designed to enhance road safety by providing real- time accident detection, analysis, and emergency response. The system integrates a vehicle unit that consists of a microcontroller (Arduino), an MPU6050 sensor for motion detection, an MQ3 gas sensor for alcohol detection, a fire sensor, a GPS system for location tracking, and a GSM module for communication. Additionally, the system utilizes the ESP8266 Wi-Fi module to enable real-time data transmission.

The vehicle unit, installed in the vehicle, continuously monitors key parameters such as speed, acceleration, impact force, environmental conditions, and driver behavior. In the event of an accident, the system automatically detects the incident and immediately triggers an alert via GSM, sending real-time notifications to emergency services, nearby hospitals, and predefined contacts, including the vehicle owner. The alert includes the exact geolocation of the accident site, ensuring rapid response and assistance. Simultaneously, all recorded data is transmitted to a cloud-based platform like ThingSpeak for secure storage, analysis, and visualization.

The system enables detailed accident reconstruction by analyzing factors such as driver behavior, vehicle dynamics, and road conditions, assisting authorities and insurance agencies in determining the cause of accidents. Additionally, the platform supports machine learning-based predictive analytics, allowing the identification of high-risk driving patterns and potential accident-prone zones. This information can be leveraged to enhance road safety policies, driver training programs, and smart traffic management systems.

Beyond accident detection, the system provides remote diagnostics and real-time vehicle health monitoring, helping to prevent mechanical failures that could lead to accidents. By integrating IoT, cloud computing, and intelligent analytics, this system aims to reduce emergency response times, improve accident investigations, enhance driver awareness, and contribute to safer road infrastructure. Ultimately, this approach fosters a data-driven, proactive strategy for accident prevention and road safety improvement.

Keywords: GSM, GPS, ThingSpeak, MPU6050 sensor, MQ3 gas sensor, fire sensor, ESP8266, Arduino.

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ACRONYM	DESCRIPTION
IoT	Internet of Things
GPS	Global Positioning System
LCD	Liquid Crytal Display
GPIO	General Purpose Input Output
IDE	Integrated Development Environment
USB	Universal Serial Bus
I/O	Input Output
V2V	Vehicle to vehicle communication
Tx	Transmitter
Rx	Receiver

CHAPTER 1
INTRODUCTION

1.1 Introduction

The Internet of Things (IoT) has emerged as a transformative technology, revolutionizing the way we connect and interact with the world around us. By interconnecting embedded computing devices within the existing internet infrastructure, IoT offers unparalleled levels of connectivity, enabling seamless communication between devices, systems, and services. This connectivity extends far beyond traditional machine-to-machine (M2M) communications, encompassing a diverse range of protocols, domains, and applications. In virtually every field of automation, IoT has paved the way for innovative solutions and advanced applications, unlocking new possibilities for efficiency, productivity, and convenience. From optimizing energy distribution through Smart Grids to revolutionizing healthcare with heart monitoring implants, IoT has become an integral part of modern life. One particularly promising application of IoT technology lies in the realm of vehicle safety. With the proliferation of smart objects and embedded sensors, it is now possible to develop sophisticated systems for accident detection and tracking. The Smart Accident Detection and Alert System is an intricate system to detect accidents and alert emergency services using Internet of Things (IoT) technology. The objective of this system is to minimize emergency response time and increase the possibility of saving lives in case of accidents The system comprises a setup of in- vehicle sensors that record data like speed, GPS location, and accelerometer readings. A central server receives these data and processes it to determine whether an accident has taken place. The sensors communicate with the server using 3G, 4G, Wi-Fi or any other such wireless networks. The server analyses the data using machine learning algorithms in order to decide if an accident has happened.If an accident is detected, the system notifies emergency services and designated contacts with details about the location and severity of the accident. The victim’s friends or family can also receive notifications from the system informing them of the accident. To send alerts, the system utilizes a variety of communication methods, including SMS, email, and push notifications.

One of the major advantages of this system is that it can detect accidents even in remote locations where cellular connectivity is unavailable. In such instances, the system sends alerts via satellite communication. The system can also communicate with existing emergency services, enabling more efficient and coordinated responses. For example, the system can automatically notify nearby hospitals to prepare for arriving patients. The system may additionally inform you how many passengers were in the vehicle, what kind of vehicle it was, and how serious the collision was. Another beneficial feature of the system is its ability to collect data on driving behaviour, which may be helpful in improving road safety. The system can acquire data on factors like speed, acceleration, and braking that can be used to spot unsafe driving behaviours. With the use of this data, drivers can receive feedback and be encouraged to adopt safer driving habits.

In order to provide additional functionality, the system can be integrated with other IoT devices. For instance, it can be connected to wearable gadgets like smart watches or health monitors to provide information regarding the victim’s health status. Emergency services can use this information to provide appropriate medical assistance.

Figure 1.1: Internet of Things

The Internet of Things can be described as connecting everyday objects like smart-phones, Internet TVs, sensors and actuators to the Internet where the devices are intelligently linked together enabling new forms of communication between things and people, and between things themselves. Building IoT has advanced significantly in the last couple of years since it has added a new dimension to the world of information and communication technologies. The Internet has come a long way over the last 30 years. Old- fashioned IPv4 is giving way to IPv6 so that every device on the Internet can have its own IP address.

1.2 What is an IoT (Internet of Things)

Let's look closely at our mobile device which contains GPS Tracking, Mobile Gyroscope, Adaptive brightness, Voice detection, Face detection etc. These components have their own individual features, but what about if these all communicate with each other to provide a better environment? For example, the phone brightness is adjusted based on my GPS location or my direction

Connecting everyday things embedded with electronics, software, and sensors to internet enabling to collect and exchange data without human interaction called as the Internet of Things (IoT).

The term "Things" in the Internet of Things refers to anything and everything in day-to-day life which is accessed or connected through the internet.

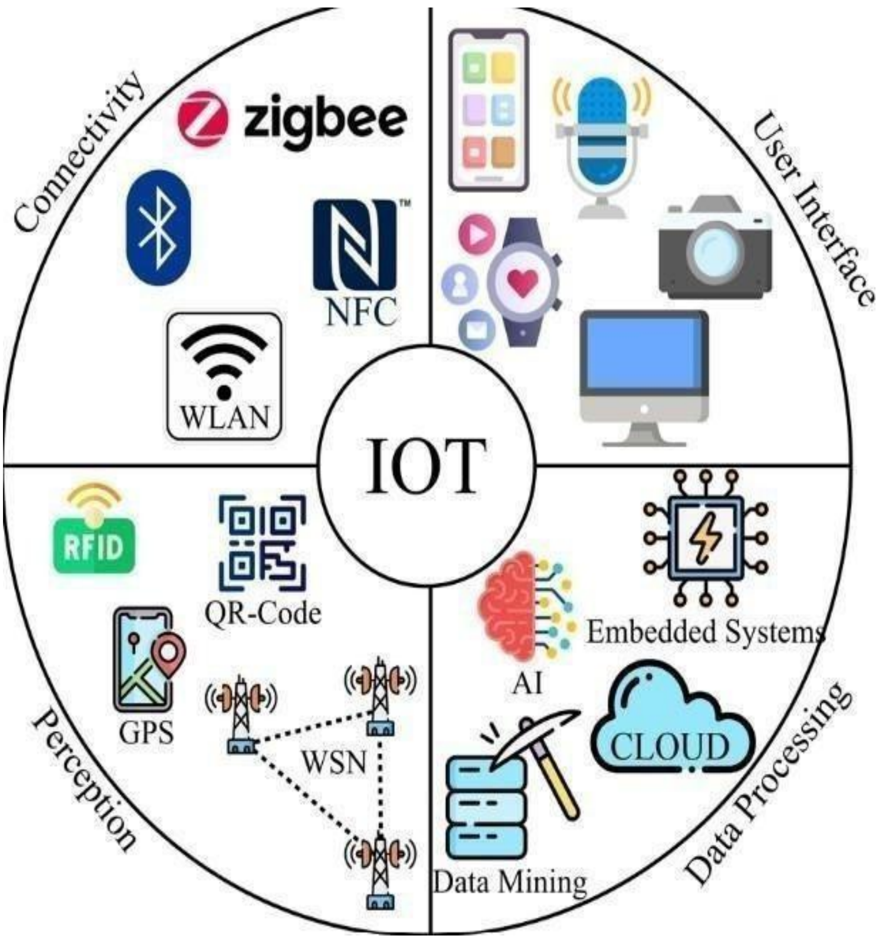

Figure 1.2: Introduction of IoT

1.3 Characteristics of IoT

- **Connectivity:** Connectivity is an important requirement of the IoT infrastructure. Things of IoT should be connected to the IoT infrastructure. Anyone, anywhere, anytime can connect, this should be guaranteed at all times. For example, the connections between people through Internet devices like mobile phones, and other gadgets, also a connection between Internet devices such as routers, gateways, sensors, etc.
- **Intelligence and Identity:** The extraction of knowledge from the generated data is very important. For example, a sensor generates data, but that data will only be useful if it is interpreted properly. Each IoT device has a unique identity. This identification is helpful in tracking the equipment and at times for querying its status.
- **Data-driven:** Data-driven is a key characteristic of the Internet of Things (IoT). IoT devices and systems collect vast amounts of data from sensors and other sources, which can be analysed and used to make data-driven decisions.
- In IoT systems, data is collected from embedded sensors, actuators, and other sources, such as cloud services, databases, and mobile devices. This data is used to gain insights into the environment, improve operational efficiency, and make informed decisions.
- **Self-Up gradation:** As we saw above, updating the software regularly is important. But who has the time to remember to do that? Thus, with its artificial intelligence, IoT upgrades itself without human help. It also allows the setup of a network for the addition of any new IoT devices. Thus, the technology can quickly start working without delay if the setup has already been done

1.4 Applications of IoT

One of the best and the most practical applications of IoT, smart homes really take both, convenience and home security, to the next level. Though there are different levels at which IoT is applied for smart homes, the best is the one that blends intelligent utility systems and entertainment together. For instance, your electricity meter with an IoT device giving you insights into your everyday water usage, your set-top box that allows you to record shows from remote, Automatic Illumination Systems, Advanced Locking Systems, Connected Surveillance

Systems all fit into this concept of smart homes.

- **Smart City**

  Not just internet access to people in a city but to the devices in it as well – that‘s what smart cities are supposed to be made of. And we can proudly say that we‘re going towards realizing this dream. Efforts are being made to incorporate connected technology into infrastructural requirements and some vital concerns like Traffic Management, Waste Management, Water Distribution, Electricity Management, and more. All these work towards eliminating some day-to-day challenges faced by people and bring in added convenience.

- **Self-driven Cars**

  We‘ve seen a lot about self-driven cars. Google tried it out, Tesla tested it, and even Uber came up with a version of self-driven cars that it later shelved. Since it‘s human Lives on the roads that we‘re dealing with, we need to ensure the technology has all that it takes to ensure better safety for the passenger and those on the roads. The cars use several sensors and embedded systems connected to the Cloud and the internet to keep generating data and sending them to the Cloud for informed decision- making through Machine Learning. Though it will take a few more years for the technology to evolve completely and for countries to amend laws and policies, what we‘re witnessing right now is one of the best applications of IoT.

- **Telehealth**

  Telehealth, or Telemedicine, hasn‘t completely flourished yet. Nonetheless, it has great future potential. IoT Examples of Telemedicine include the digital communications.

- **Smart Supply-chain Management**

  Supply-chains have stuck around in the market for a while now. A common example can be Solutions for tracking goods while they are on the road. Backed with IoT technology, they are sure to stay in the market for the long run.

- **Farming**

  Farming is one sector that will benefit the most from the Internet of Things. With so many developments happening on tools farmers can use for agriculture, the future is sure promising. Tools are being developed for Drip Irrigation, understanding crop, Water Distribution, drones for Farm Surveillance, and more. These will allow farmers to come up with a more productive yield and take care of the concerns better.

- **Wearable**
  Wearables remain a hot topic in the market, even today. These devices serve a wide range of purposes ranging from medical, wellness to fitness. Of all the IoT startups, Jawbone, a wearables maker, is second to none in terms of funding.

1.5 Architecture Components of IoT

The hardware unit in an IoT system falls into one or more of the following categories:

- Processing units Storage unit
- Communication unit
- A person or some kind of active digital entity (e.g., a service, an application or a software agent) that has a goal. The attainment of the goal is achieved via interaction with the physical environment. This interaction is mediated by the IoT.

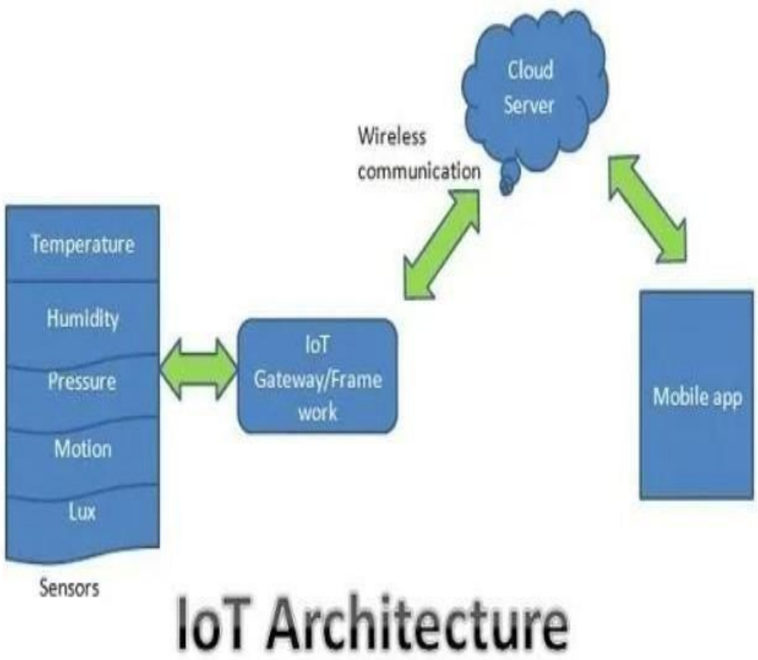

Figure 1.5: Architecture of IoT

1.6 Advantages of IoT

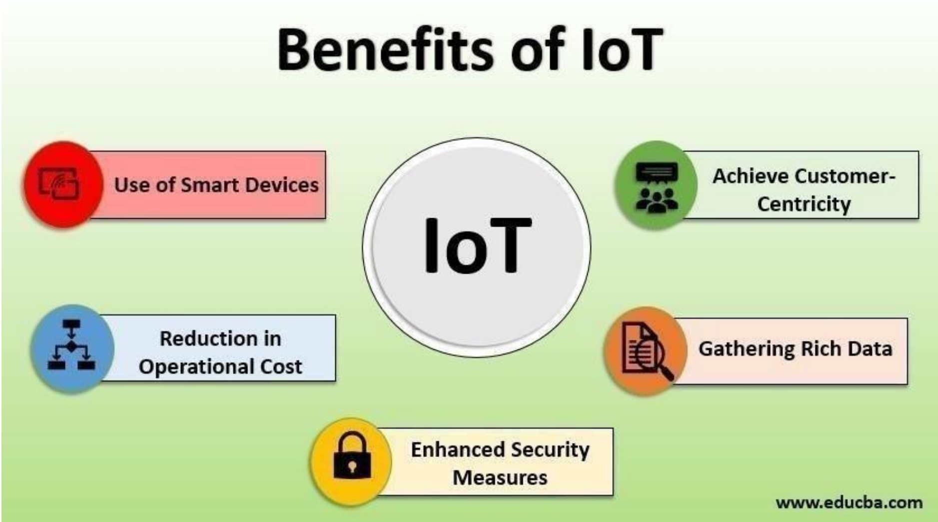

Figure 1.6: Benefits of IoT

1.6.1 Technical Optimization

IoT technology helps a lot in improving technologies and making them better. Example, with IoT, a manufacturer is able to collect data from various car sensors. The manufacturer analyses them to improve its design and make them more efficient.

1.6.2 Reduced Waste

IoT offers real-time information leading to effective decision making & management of resources. For example, if a manufacturer finds an issue in multiple car engines, he can track the manufacturing plan of those engines and solves this issue with the manufacturing belt.

1.6.3 Improved Customer Engagement:

IoT allows you to improve customer experience by detecting problems and improving the process.

1.7 Disadvantages of IoT:

1.7.1 Security

IoT technology creates an ecosystem of connected devices. However, during this process, the system may offer little authentication control despite sufficient security measures.

1.7.2 Privacy

The use of IoT, exposes a substantial amount of personal data, in extreme detail, without the user's active participation. This creates lots of privacy issues.

1.7.3 Flexibility

There is a huge concern regarding the flexibility of an IoT system. It is mainly regarding integrating with another system as there are many diverse systems involved in the process.

1.7.5 Complexity:

- The design of the IoT system is also quite complicated.
- Moreover, it's deployment and maintenance also not very easy.

1.7.6 Compliance:

IoT has its own set of rules and regulations. However, because of its complexity, the task of compliance is quite challenging.

1.8 Problem Statement

Drunk driving is a leading cause of road accidents, often resulting in severe injuries and fatalities, while delays in emergency response can further escalate the consequences. This project proposes an intelligent vehicle safety system that detects driver intoxication and prevents the vehicle from starting. In the event of a collision or fire, the system automatically sends real-time GPS coordinates to emergency contacts, ensuring a swift response. By incorporating these features, the system aims to enhance road safety, reduce drunk driving incidents, and facilitate timely assistance during emergencies.

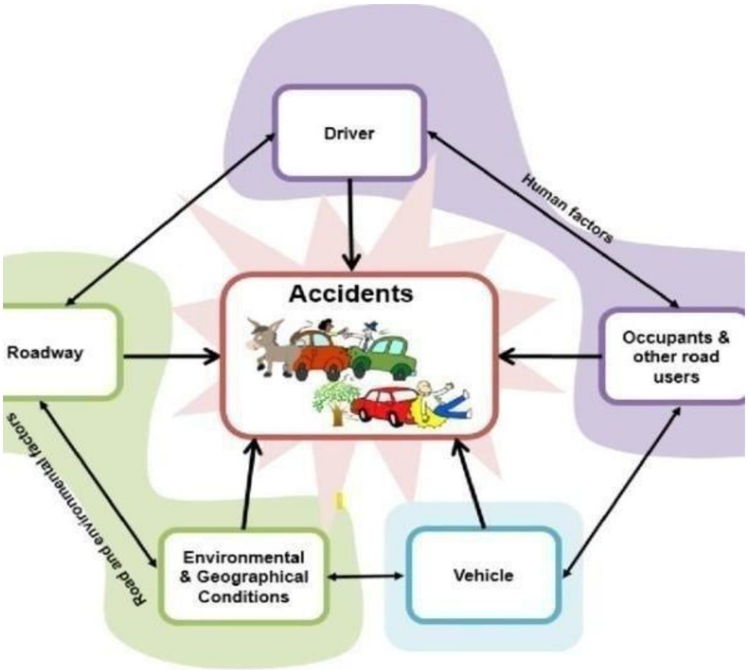

Figure 1.8: Causes of accidents

1.9 Motivation

The motivation behind our work is, road accidents are a major cause of fatalities and injuries worldwide, often due to human error, poor road conditions, or mechanical failures. Delayed emergency response worsens outcomes, as traditional reporting relies on bystanders or vehicle occupants, which may not always be possible. This project aims to address these issues by developing a real-time accident monitoring system using IoT for instant detection, automated alerts, and real-time data transmission. Integrating sensors, microcontrollers, and wireless communication ensures faster emergency response, accurate location tracking, and detailed accident analysis. The system also detects hazards like alcohol influence and fire outbreaks, while remote vehicle health monitoring promotes safer driving and proactive maintenance, contributing to a more secure and connected transportation system.

CHAPTER 2

EMBEDDED SYSTEM

2.1 Embedded Systems

An embedded system is a computer system designed to perform one or a few dedicated functions often with real-time computing constraints. It is embedded as part of a complete device often including hardware and mechanical parts. By contrast, a general-purpose computer, such as a personal computer (PC), is designed to be flexible and to meet a wide range of end-user needs. Embedded systems control many devices in common use today.

Embedded systems are controlled by one or more main processing cores that are typically either microcontrollers or digital signal processors (DSP). The key characteristic, however, is being dedicated to handle a particular task, which may require very powerful processors. For example, air traffic control systems may usefully be viewed as embedded, even though they involve mainframe computers and dedicated regional and national networks between airports and radar sites. (Each radar probably includes one or more embedded systems of its own.)

Since the embedded system is dedicated to specific tasks, design engineers can optimize it to reduce the size and cost of the product and increase the reliability and performance. Some embedded systems are mass-produced, benefiting from economies of scale.

Physically embedded systems range from portable devices such as digital watches and MP3 players, to large stationary installations like traffic lights, factory controllers, or the systems controlling nuclear power plants. Complexity varies from low, with a single microcontroller chip, to very high with multiple units, peripherals and networks mounted inside a large chassis or enclosure.

In general, "embedded system" is not a strictly definable term, as most systems have some element of extensibility or programmability. For example, handheld computers share some elements with embedded systems such as the operating systems and microprocessors which power them, but they allow different applications to be loaded and peripherals to be connected. Moreover, even systems which don't expose programmability as a primary feature generally need to support software updates. On a continuum from "general purpose" to "embedded", large application systems will have subcomponents at most points even if the system as a whole is "designed to perform one or a few dedicated functions", and is thus appropriate to call "embedded".

In many ways, programming for an embedded system is like programming PC 15 years ago. The hardware for the system is usually chosen to make the device as cheap as possible. Spending an extra dollar a unit in order to make things easier to program can cost millions. Hiring a programmer for an extra month is cheap in comparison. This means the programmer must make do with slow processors and low memory, while at the same time battling a need for efficiency not seen in most PC applications. Below is a list of issues specific to the embedded field.

2.2 Resources:

To save costs, embedded systems frequently have the cheapest processors that can do the job. This means your programs need to be written as efficiently as possible. When dealing with large data sets, issues like memory cache misses that never matter in PC programming can hurt you. Luckily, this won't happen too often- use reasonably efficient algorithms to start, and optimize only when necessary. Of course, normal profilers won't work well, due to the same reason debuggers don't work well.

Memory is also an issue. For the same cost savings reasons, embedded systems usually have the least memory they can get away with. That means their algorithms must be memory efficient (unlike in PC programs, you will frequently sacrifice processor time for memory, rather than the reverse). It also means you can't afford to leak memory. Embedded applications generally use deterministic memory techniques and avoid the default "new" and "malloc" functions, so that leaks can be found and eliminated more easily. Other resources programmers expect may not even exist. For example, most embedded processors do not have hardware FPUs (Floating-Point Processing Unit). These resources either need to be emulated in software, or avoided altogether.

2.3 Real Time Issues:

Embedded systems frequently control hardware, and must be able to respond to them in real time. Failure to do so could cause inaccuracy in measurements, or even damage hardware such as motors. This is made even more difficult by the lack of resources available. Almost all embedded systems need to be able to prioritize some tasks over others, and to be able to put off/skip low priority tasks such as UI in favor of high priority tasks like hardware control. Embedded systems which are used to perform a specific task or operation in a specific time period those systems are called as real-time embedded systems. There are two types of real-time embedded systems.

- Hard Real-time embedded systems:

These embedded systems follow an absolute dead line time period i.e.., if the tasking is not done in a particular time period, then there is a cause of damage to the entire equipment. Eg: consider a system in which we have to open a valve within 30 milliseconds. If this valve is not opened in 30 ms this may cause damage to the entire equipment. So in such cases we use embedded systems for doing automatic operations.

- Soft Real Time embedded systems:

These embedded systems follow a relative dead line time period i.e.., if the task is not done in a particular time that will not cause damage to the equipment.

Eg: Consider a TV remote control system, if the remote control takes a few milliseconds delay it will not cause damage either to the TV or to the remote control. These systems which will not cause damage when they are not operated at considerable time period those systems come under soft real-time embedded systems.

2.3.1 Different types of processing units:

The central processing unit (C.P.U.) can be any one of the following microprocessors, micro controller, digital signal processing.

2.3.1.1 Among these Microcontroller is of low-cost processor and one of the main advantage of microcontrollers is, the components such as memory, serial communication interfaces, analog to digital converters etc.., all these are built on a single chip. The numbers of external components that are connected to it are very less according to the application.

2.3.1.2 Microprocessors are more powerful than microcontrollers. They are used in major applications with a number of tasking requirements. But the microprocessor requires many external components like memory, serial communication, hard disk, input output ports etc.., so the power consumption is also very high when compared to microcontrollers.

2.3.1.3 Digital signal processing is used mainly for the applications that particularly involved with processing of signals

2.4 Applications of Embedded Systems

2.4.1 Consumer applications

At home we use a number of embedded systems which include microwave oven, remote control, VCD players, DVD players, Camera etc.

2.4.2 Industrial automation

Today a lot of industries are using embedded systems for process control. In industries we design the embedded systems to perform a specific operation like monitoring temperature, pressure, humidity, voltage, current etc.., and basing on these monitored levels we do control other devices, we can send information to a centralized monitoring station. In critical industries where human presence is avoided there, we can use robots which are programmed to do a specific operation.

2.5 Block Diagram of Embedded System

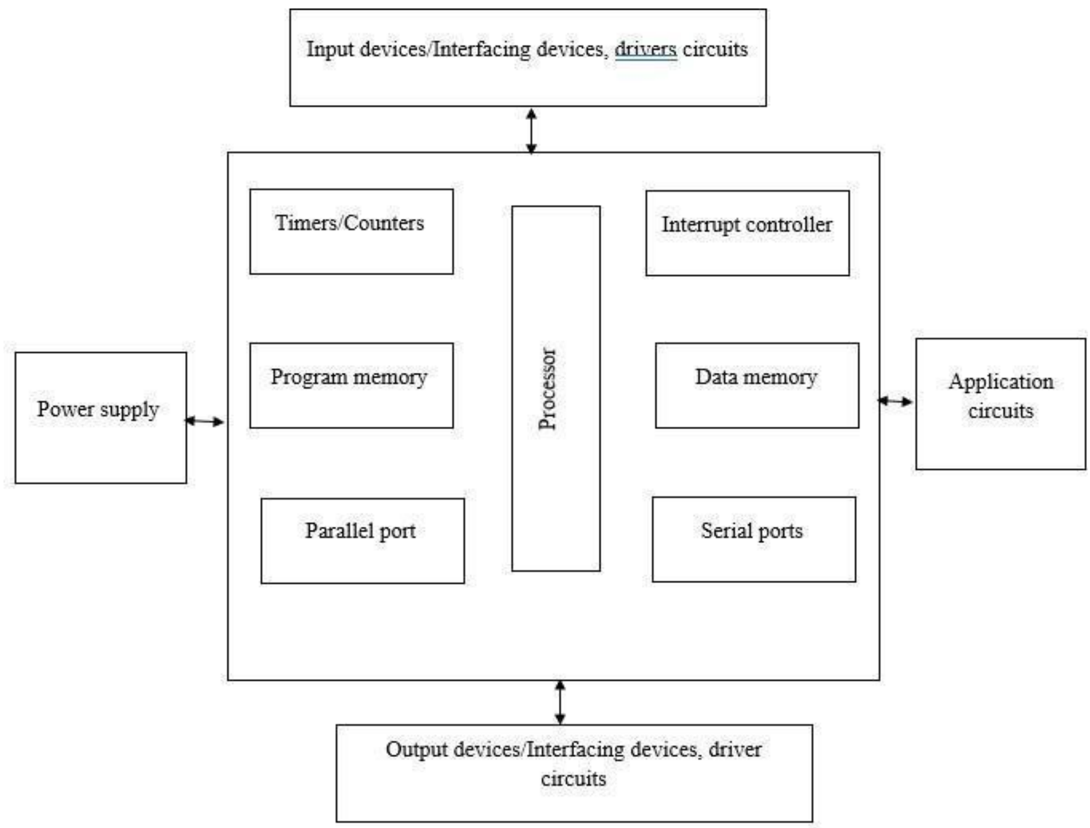

Figure 2.5: Block Diagram of Embedded System

The block diagram of a typical embedded system, which is a dedicated computing system designed to perform specific tasks within a larger mechanical or electrical system. At the core of the embedded system is the processor, which executes instructions and controls the overall operation. The program memory stores the firmware or software necessary for the system's functioning, while the data memory holds temporary data and variables required during execution. Supporting the processor are essential components like the timer/counter, used for timing operations and event counting, and the interrupt controller, which manages interrupt signals and ensures the processor responds promptly to high-priority events.

The system also includes serial ports and a parallel port for communication with other devices, either serially or in parallel. These ports allow data exchange between the embedded system and external peripherals. The embedded system interfaces with the external world through input devices and output devices, which are connected via driver and interfacing circuits. Input devices like sensors provide real-time data to the processor, while output devices such as actuators and displays execute the system's responses. Additionally, application-specific circuits are connected to the system to carry out specialized tasks, depending on the system's intended use. The entire system is powered by a power supply, which ensures all components receive the necessary electrical energy to function properly. Overall, this architecture allows the embedded system to perform its tasks efficiently, reliably, and often in real-time.

CHAPTER 3
LITERATURE SURVEY

An IoT based Vehicle Accident Prevention and Tracking System for Night Drivers by Aishwarya S.R, Ashish Rai, Charitha, Prasanth M.A, and Savitha S.C [1]

Road accidents are increasing, and timely medical aid can save lives. Our vehicle accident detection and notification system reduces the delay between an accident and the arrival of help. When an accident occurs, contact sensors on the vehicle send a signal to the ESP32 microcontroller, which then triggers a GSM alert: "Accident Occurred!" The alert, along with the vehicle's live GPS location, is sent to a mobile app. In case of theft, the owner can also track the vehicle in real-time and remotely turn off the ignition or activate a buzzer through the app. This system aims to ensure quick medical response and enhance vehicle security.

Smart Helmet – Intelligent Safety for Motorcyclist using Raspberry Pi and OpenCV by Sadhana B Shabrin, Bhagyashree Jagadish Nikharge, Maithri M Poojary, and T Pooja [2]

Smart Helmet - Intelligent Safety Helmet for Motorcyclist is a project undertaken to increase the rate of road safety among motorcyclists. The idea is obtained after knowing that there is increased number of fatal road accidents over the years. Through the study identified, it is analysed that the helmets used is not in safety features such as not wearing a helmet string and not use the appropriate size. Therefore, this project is designed to introduce safety systems for the motorcyclist to wear the helmet properly. With the use of Image processing unit using Raspberry Pi and Open Cv , the motorcycle can move if there is helmet pound wearing, in accordance with the project title Smart Helmet - Intelligent Safety for Motorcyclist using Raspberry Pi and Open Cv. Safety system applied in this project meet the characteristics of a perfect rider and the application should be highlighted. The project is expected to improve safety.

Wireless System for Vehicle Accident Detection and Reporting using Accelerometer and GPS by Shailesh Bhavthankar and Prof. H.G. Sayyed [3]

Accident threatens human lives more and mainly road accident is common today. During accident many people lose their life because medical services and family member not getting accidental information on time. In this paper, an efficient vehicle wireless system is designed and implemented for vehicle accident detection and reporting using accelerometer and GPS. Accelerometer sensor is used to detect crash and GPS give location of vehicle. In case of any accident, the system send automated message to the preprogrammed number such as family member or emergency medical services via GSM.

Advanced Embedded System of Vehicle Accident Detection and Tracking System by Sarika R. Gujar and Prof. A. R. Itkikar [4]

Infrastructure and also technological advancements have actually made life less complex for individuals, causing huge demand for vehicles. This results in an increase in road mishaps. An accident is a vulnerable and unintended occasion, as well as it can additionally occur because of the carelessness of the vehicle drivers. With the increasing number of road accidents and the high mortality rate associated with them, there is a growing need for a system that can detect accidents and provide timely assistance to the victims. In this project, we propose an IoT based Accident Detection and Rescue System that leverages various sensors, GSM, GPS, fire sensor, IR sensor, MQ135 Gas sensor, I2C LCD, DC Motor modules, to detect an accident and alert the concerned authorities. The proposed system has the potential to reduce the response time in the event of an accident, thereby improving the chances of survival for the victim. The project combines the latest advancements in IoT technology and Thingspeak to store the data.

Load Balancing Under Heavy Traffic in RPL Routing Protocol for Low Power and Lossy Networks by Hyung-Sin Kim, Hongchan Kim, Jeongyeup Paek, and Saewoong Bahk [5]

RPL is an IPv6 routing protocol for low-power and lossy networks (LLNs), designed for various applications like smart grid AMIs, industrial and environmental monitoring, and wireless sensor networks. It enables bi-directional IPv6 communication on resource-constrained LLN devices, supporting the Internet of Things (IoT) through multihop mesh networks. This article investigates RPL's load balancing and congestion issues, showing that most packet losses under heavy traffic stem from congestion and poor routing parent selection. To address this, a queue utilization-based RPL (QU-RPL) is proposed, which significantly enhances end-to-end packet delivery by considering both queue utilization and hop distance to the LLN border router (LBR) during parent selection. Thanks to its load balancing ability, QU-RPL effectively reduces queue losses and boosts packet delivery. Implemented on a low-power embedded platform, QU-RPL is validated through real-world tests over IEEE 802.15.4. Results show it reduces queue loss by up to 84% and improves packet delivery ratio by up to 147% over standard RPL.

CHAPTER 4

EXISTING SYSTEM

People in general or crisis administrations don't have the foggiest idea about the close-by area and the level of administrations. The absence of such data may cause a few causalities. Consequently, the exploration inquiry emerges in a manner to answer how to run the crisis vehicle to achieve the mishap spot in correct time. Thus, directing the vehicle and making movement sign control according to vehicle development is important. There is death toll because of the postponement in the entry of Emergency vehicle to the mischance place. To accomplish these obliged conditions we propose executing a System called Emergency Vehicles Monitoring System (EVMS) Emergency vehicle checking System can incorporate GPS, GSM, GIS administrations. The GPS gadgets convey the information concerning the Emergency vehicle position and their briefest course. The primary server the Google guide permits to pick briefest course between crisis vehicle and mischance spot.

It would be of extraordinary utilization to the crisis vehicle if the traffic signals in the way of the mishap spot are ON. Accordingly, we propose another outline for consequently controlling the activity signals and emergency vehicle checking attaining the aforementioned errand so that the crisis vehicle would have the capacity to cross all the traffic control signal without holding up. Each movement intersection will have a controller controlling the activity flow. The movement control is alluded to as knobs and every knob will have a GSM (Global System for portable correspondence) modem connected to the controller. The hubs are controlled by a main server by sending the control messages to their GSM (Global system for portable correspondence) modems. Nowadays the use of vehicles increases in the proportion of the population. Due to the traffic congestion, the accidents are also increasing day by day. This causes the loss of life due to the delay in the arrival of ambulances to the accident spot or from the accident spot to the hospital and save their lives. So, it is necessary to take the accident victim to the hospital and provide medical services as soon as possible. Whenever an accident occurs, it has to be informed to the investigation unit. So, it is also beneficial if the intimation is reached to the inquiry section so that the time for the investigation can be minimized.

Limitations of existing system:

- Lack of real-time information about nearby emergency services can cause delays in medical assistance, increasing the risk of fatalities.
- Inefficient route planning and traffic congestion hinder emergency vehicles from reaching accident locations on time.
- Traffic signals are not dynamically controlled to prioritize emergency vehicles, leading to unnecessary delays at intersections.
- The high volume of vehicles on roads increases traffic congestion, making accident response and hospital transportation slower.
- The absence of an automated system to inform investigation units about accidents delays legal and insurance processes.
- Integrating GPS, GSM, and GIS for efficient route optimization is challenging and requires advanced infrastructure.
- Dependence on mobile networks for communication may result in failures in remote or low-network areas.
- The system's effectiveness depends on the widespread adoption of smart traffic management and emergency vehicle tracking technologies.

CHAPTER 5
PROPOSED SYSTEM

This work proposes Accident Detection Rescue System using IoT. IoT is very useful in these specters it replaces the conventional monitoring systems with a more efficient scheme.

5.1 Block Diagram

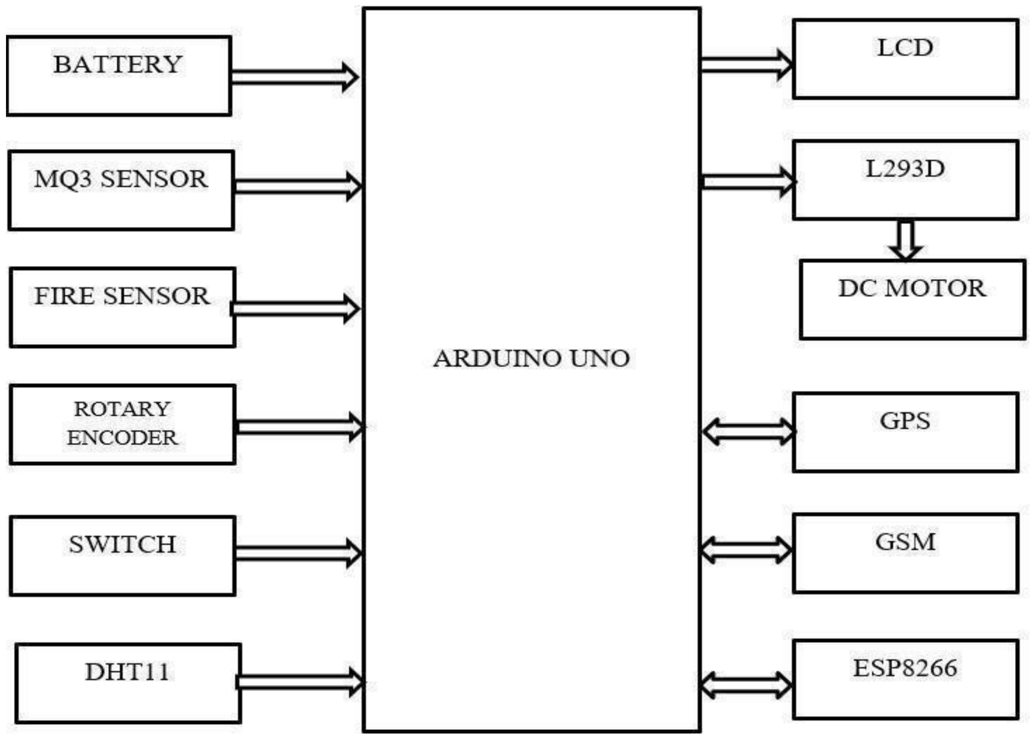

Figure 5.1: Block diagram of proposed system

The proposed system consists of a MPU6050 sensor, MQ3 gas sensor, Fire sensor, GPS system, GSM module if any accident occurred, then GSM module sends the accident location to the t emergency contacts. The mpu6050 sensor used in the vehicle will continuously sense for any large change in the vehicle. The sensed data is given to the controller. GPS module finds out the current position of the vehicle which is the location of the accident and gives that data to the GSM module. The GSM module sends this notification to the whose GSM number is already there in the module as an emergency number. Alcohol sensor (MQ-3) to determine the driver's breath alcohol content and prevent the ignition from starting in the first place. A flame sensor detectsthe presence of a fire or flame that occurs in a car.

In emergencies, the microcontroller activates the braking system to apply brakes swiftly and decisively, enhancing safety. Anti-collision braking system (ACBS) is a safety system that can identify when a possible collision is about to occur and responds by autonomously activating the brakes to slow a vehicle prior to impact or bring it to a stop to avoid a collision. Performing a car brake safety system using switch and designing a car that is less sensitive to driving.

A rotary encoder can calculate a vehicle's speed by attaching it to the wheel hub, where each rotation of the wheel generates a set of pulses from the encoder, allowing you to count the number of pulses over a specific time period, which when converted using the wheel circumference, gives you the vehicle's speed; essentially, the faster the wheel rotates, the higher the pulse frequency, indicating a higher speed.

A DHT11 sensor is used to measure the temperature inside a vehicle by providing a digital reading of the surrounding air temperature. All the sensor value are updated to the thingspeak server.

5.2 Arduino UNO

The Arduino UNO is a widely used open-source microcontroller board based on the Microchip ATmega328P microcontroller and developed by Arduino. The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits The board features 14 Digital pins and 6 Analog pins. It is programmable with the Arduino IDE(Integrated Development Environment) via a type B USB cable. It can be powered by a USB cable or by an external 9v battery, though it accepts voltages between 7 and 20 volts. It is also similar to the Arduino Nano and Leonardo. The hardware reference design is distributed under a Creative Commons Attribution Share-Alike 2.5 license and is available on the Arduino website. Layout and production files for some versions of the hardware are also available. "Uno" means one in Italian and was chosen to mark the release of Arduino Software (IDE) 1.0 The Uno board and version 1.0 of Arduino Software (IDE) were the reference versions of Arduino, now evolved to newer release. The Uno board is the first in a series of USB Arduino boards, and the reference model for the Arduino platform. The ATmega328 on the Arduino Uno comes preprogrammed with a bootloader that allows uploading new code to it without the use of an external hardware programmer.

TECHNICAL SPECIFICATIONS

- Microcontroller: Microchip ATmega328P
- Operating Voltage: 5 Volt
- Input Voltage: 7 to 20 Volts
- Digital I/O Pins: 14 (of which 6 provide PWM output)
- Analog Input Pins: 6
- DC Current per I/O Pin: 20 mA
- DC Current for 3.3V Pin: 50 mA
- Flash Memory: 32 KB of which 0.5 KB used by bootloader
- SRAM: 2 KEEPROM: 1 KB
- Clock Speed: 16 MHz
- Length: 68.6 mm
- Width: 53.4 mm
- Weight: 25

➢ It communicates using the original STK500 protocol. The Uno also differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega16U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter. The Arduino Uno is facilitated by the user- friendly Arduino Integrated Development Environment (IDE), enabling seamless code writing, compiling, and uploading. With a rich library of functions and example code, users can swiftly develop projects and interface with components. Powering options are flexible, accommodating USB, external supplies, or batteries, supported by an onboard voltage regulator for stability. Complementing its core features, the Arduino Uno boasts compatibility with a plethora of shields, expanding its capabilities for wireless communication, motor control, and displays.

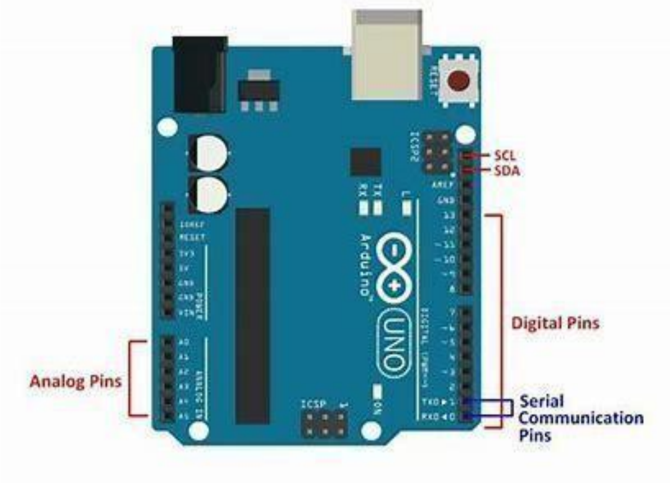

Figure 5.2.1: Arduino UNO

GENERAL PIN FUNCTIONS

• LED: There is a built-in LED driven by digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.

• VIN: The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.

• 5V: This pin outputs a regulated 5V from the regulator on the board. The board can be supplied with power either from the DC power jack (7 - 20V), the USB connector (5V), or the VIN pin of the board (7-20V). Supplying voltage via the 5V or 3.3V pins bypasses the regulator and can damage the board.

• 3V3: A 3.3-volt supply generated by the on-board regulator. Maximum current draw is 50 mA.

• GND: Ground pins.

• IOREF: This pin on the Arduino board provides the voltage reference with which the microcontroller operates. A properly configured shield can read the IOREF pin voltage and select the appropriate power source or enable voltage translators on the outputs to work with the 5V or 3.3V.

• Serial: pins 0 (RX) and 1 (TX). Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to- TTL Serial chip.

• External Interrupts: pins 2 and 3. These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value.

5.3 Battery

Batteries serve as vital energy sources within IoT-based anti-poaching and fire detection systems for valuable trees, providing the necessary power for sensor operation, data transmission, and communication in remote forest environments. Their role as independent power sources is particularly crucial in areas where grid power access is limited or non-existent. The energy storage capacity of batteries directly influences the duration of system operation without requiring recharging or replacement, making it imperative to select batteries with adequate capacity to sustain system functionality over extended periods. Furthermore, the choice of battery type, such as lithium-ion, alkaline, or rechargeable nickel-metal hydride, depends on factors like energy density, lifespan, and operating temperature range, with consideration for environmental conditions prevalent in forest ecosystems. Rechargeable batteries offer the advantage of reusability, reducing the need for frequent replacements, albeit requiring charging infrastructure and careful monitoring of cycle life. Maintenance practices, including monitoring battery health, voltage levels, and safety considerations, are essential for ensuring optimal performance and mitigating risks associated with short circuits or overheating, particularly crucial in forest environments where the risk of fire may be heightened. Therefore, meticulous battery selection, installation, and maintenance are paramount to sustaining the reliability, longevity, and safety of IoT-based systems deployed for protecting valuable trees against poaching and fire threats.

5.3.1 POWER USB

Arduino board is connected to the laptop via USB cable and it is helpful for connecting and operating in an easy way.

5.4 LCD (Liquid Crystal Display)

A liquid crystal display (LCD) is a thin, flat display device made up of any number of color or monochrome pixels arrayed in front of a light source or reflector. Each pixel consists of a column of liquid crystal molecules suspended between two transparent electrodes, and two polarizing filters, the axes of polarity of which are perpendicular to each other. Without the liquid crystals between them, light passing through one would be blocked by the other. The liquid crystal twists the polarization of light entering one filter to allow it to pass through the program must interact with the outside world using input and output devices that communicate directly with a human being. One of the most common devices attached to a controller is an LCD display. Some of the most common LCDs connected to the controller are 16X1, 16x2 and 20x2 displays. This means 16 characters per line by 1 line 16 characters per line by 2 lines and 20 characters per line by 2 lines, respectively. Many microcontroller devices use 'smart LCD' displays to output visual information. LCD displays designed around LCD NT-C1611 module, are inexpensive, easy to use, and it is even possible to produce a readout using the 5X7 dots plus cursor of the display. They have a standard ASCII set of characters and mathematical symbols. For an 8-bit data bus, the display requires a +5V supply plus 10 I/O lines (RS RW D7 D6 D5 D4 D3 D2 D1 D0). For a 4-bit data bus it only requires the supply lines plus 6 extra lines (RS RW D7 D6 D5 D4). When the LCD display is not enabled, data lines are tri-state and they do not interfere with the operation of the microcontroller.

Data can be placed at any location on the LCD. For 16×1 LCD, the address locations are shown below:

Even limited to character based on modules, there is still a wide variety of shapes and sizes available. Line lengths of 8,16,20,24,32 and 40 characters are all standard, in one, two, three, four lines versions.

Several different LC technologies exist. “supertwist” types, for example, offer Improved contrast and viewing angle over the older “twisted nematic” types. Some modules are available with back lighting, so that they can be viewed in dimly-lit conditions. The back lighting may be either “electro-luminescent”, requiring a high voltage inverter circuit, or simple LED illumination.

POSITION		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
ADDRESS	LINE1	00	01	02	03	04	05	06	07	40	41	42	43	44	45	46	47

Figure 5.4.1: Address locations for a 1x16 line LCD

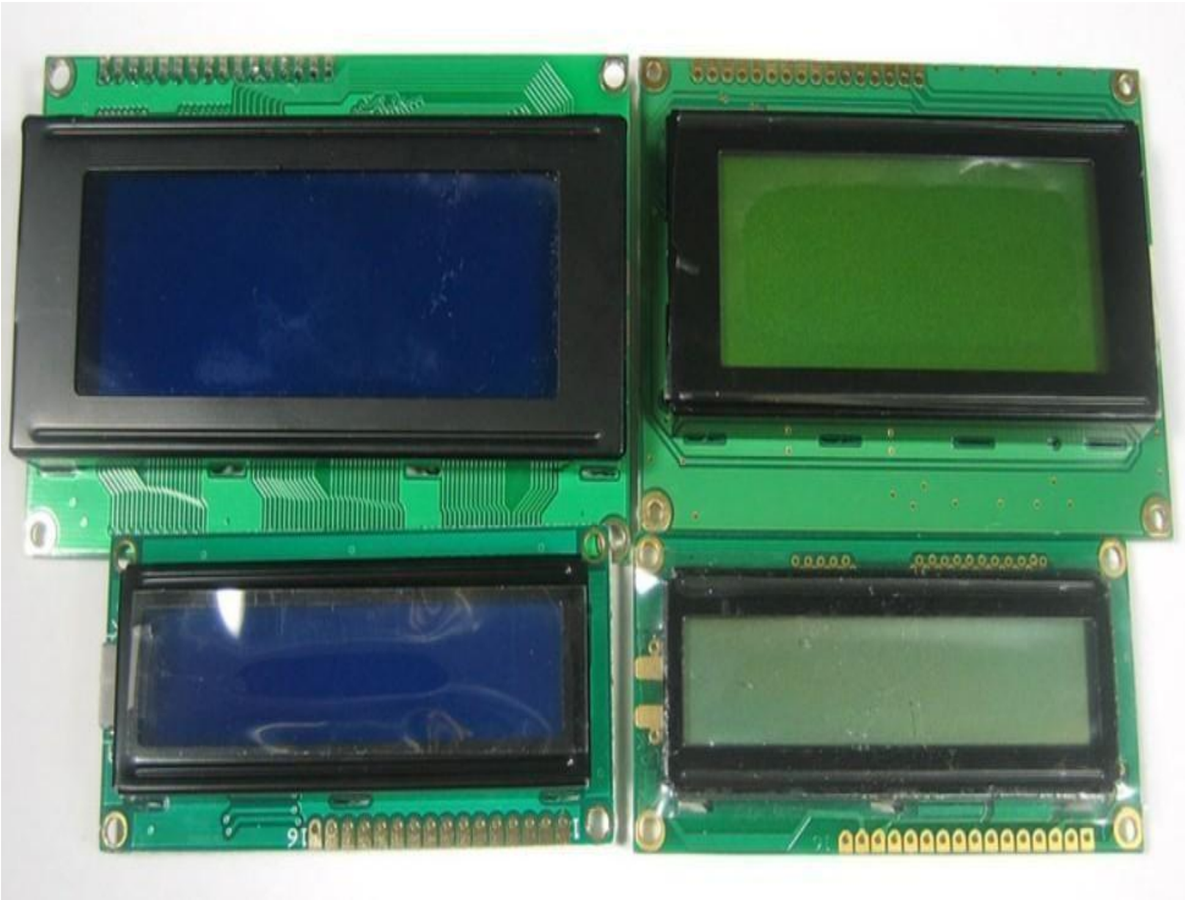

Figure 5.4.2: LCD shapes and sizes

Pin Description:

Most LCDs with 1 controller as 14 Pins and LCDs with 2 controller has 16 Pins (two pins are extra in both for back-light LED connections).

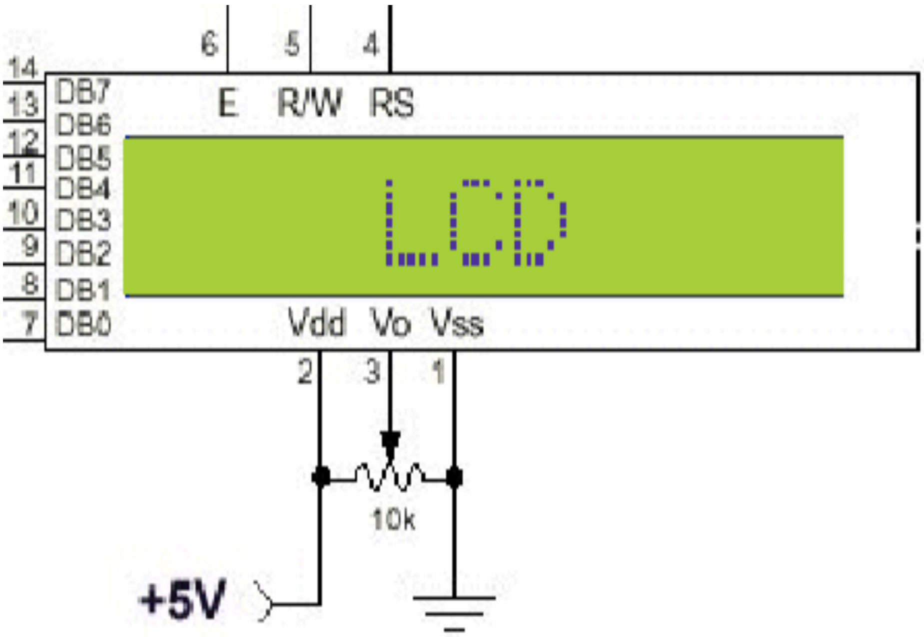

Figure 5.4.3: pin diagram of 1x16 lines lcd

PIN	SYMBOL	FUNCTION
1	Vss	Power Supply(GND)
2	Vdd	Power Supply(+5V)
3	Vo	Contrast Adjust
4	RS	Instruction/Data Register Select
5	R/W	Data Bus Line
6	E	Enable Signal
7-14	DB0-DB7	Data Bus Line
15	A	Power Supply for LED B/L(+)
16	K	Power Supply for LED B/L(-)

Control Lines:

EN:

Line is called "Enable." This control line is used to tell the LCD that you are sending it data. To send data to the LCD, your program should make sure this line is low (0) and then set the other two control lines and/or put data on the data bus. When the other lines are completely ready, bring EN high (1) and wait for the minimum amount of time required by the LCD datasheet (this varies from LCD to LCD), and end by bringing it low (0) again.

RS:

Line is the "Register Select" line. When RS is low (0), the data is to be treated as a command or special instruction (such as clear screen, position cursor, etc.). When RS is high (1), the data being sent is text data which sould be displayed on the screen. For example, to display the letter "T" on the screen you would set RS high.

RW:

Line is the "Read/Write" control line. When RW is low (0), the information on the data bus is being written to the LCD. When RW is high (1), the program is effectively querying (or reading) the LCD. Only one instruction ("Get LCD status") is a read command. All others are writes commands, so RW will almost always be low.

Finally, the data bus consists of 4 or 8 lines (depending on the mode of operation selected by the user). In the case of an 8-bit data bus, the lines are referred to as DB0, DB1, DB2, DB3, DB4, DB5, DB6, and DB7.

Logic status on control lines:

• E - 0 Access to LCD disabled

- 1 Access to LCD enabled

• R/W - 0 Writing data to LCD

- 1 Reading data from LCD

• RS - 0 Instructions

- 1 Character

Writing data to the LCD:

1) Set R/W bit to low

2) Set RS bit to logic 0 or 1 (instruction or character)

3) Set data to data lines (if it is writing)

4) Set E line to high

Read data from data lines (if it is reading) on LCD:

- Set R/W bit to high
- Set RS bit to logic 0 or 1 (instruction or character)
- Set data to data lines (if it is writing)
- Set E line to high
- Set E line to low
- **Entering Text:**

First, a little tip: it is manually a lot easier to enter characters and commands in hexadecimal rather than binary (although, of course, you will need to translate commands from binary couple of sub-miniature hexadecimal rotary switches is a simple matter, although a little bit into hex so that you know which bits you are setting). Replacing the d.i.l. switch pack with a of re-wiring is necessary.

The switches must be the type where On = 0, so that when they are turned to the zero position, all four outputs are shorted to the common pin, and in position "F", all four outputs are open circuit.

All the available characters that are built into the module are shown in Table 3. Studying the table, you will see that codes associated with the characters are quoted in binary and hexadecimal, most significant bits ("left-hand" four bits) across the top, and least significant bits ("right-hand" four bits) down the left.

Most of the characters conform to the ASCII standard, although the Japanese and Greek characters (and a few other things) are obvious exceptions. Since these intelligent modules were designed in the "Land of the Rising Sun," it seems only fair that their Katakana phonetic symbols should also be incorporated. The more extensive Kanji character set, which the Japanese share with the Chinese, consisting of several thousand different characters, is not included!

Using the switches, of whatever type, and referring to Table 3, enter a few characters onto the display, both letters and numbers. The RS switch (S10) must be "up" (logic 1) when sending the characters, and switch E (S9) must be pressed for each of them. Thus the operational order is: set RS high, enter character, trigger E, leave RS high, enter another character, trigger E, and so on.

The first 16 codes in Table 3, 00000000 to 00001111, ($00 to $0F) refer to the CGRAM. This is the Character Generator RAM (random access memory), which can be used to hold user-defined graphics characters. This is where these modules really start to show their potential, offering such capabilities as bar graphs, flashing symbols, even animated characters. Before the user-defined characters are set up, these codes will just bring up strange looking symbols.

Codes 00010000 to 00011111 ($10 to $1F) are not used and just display blank characters. ASCII codes “proper” start at 00100000 ($20) and end with 01111111 ($7F). Codes 10000000 to 10011111 ($80 to $9F) are not used, and 10100000 to 11011111 ($A0 to $DF) are the Japanese characters.

5.5 Fire sensor

A flame-sensor is one kind of detector which is mainly designed for detecting as well as responding to the occurrence of a fire or flame. The flame detection response can depend on its fitting. It includes an alarm system, a natural gas line, propane & a fire suppression system. This sensor is used in industrial boilers. The main function of this is to give authentication whether the boiler is properly working or not. The response of these sensors is faster as well as more accurate compare with a heat/smoke detector because of its mechanism while detecting the flame.

Figure 5.5: Fire sensor

5.6 Alcohol Sensor

The MQ-3 alcohol sensor is a gas sensor module specifically designed to detect the presence of alcohol vapours in the air. It utilizes a tin oxide semiconductor material that changes its electrical conductivity in the presence of alcohol molecules. When alcohol vapors come into contact with the sensor, they cause a change in its resistance, which is then measured by the sensor's electronic circuitry. The MQ-3 sensor is commonly used in various applications, including breathalyzer devices, automotive safety systems, and alcohol detection systems. It offers advantages such as high sensitivity, fast response time, and low power consumption. However, it is important to note that the MQ-3 sensor may also respond to other volatile organic compounds (VOCs) present in the environment, which can affect its accuracy in alcohol detection.

5.7 GSM (Global System for Mobile communications)

GSM (Global System for Mobile communications) is a cellular network, which means that mobile phones connect to it by searching for cells in the immediate vicinity. GSM networks operate in four different frequency ranges. Most GSM networks operate in the 900 MHz or 1800 MHz bands. Some countries in the Americas use the 850 MHz and 1900 MHz bands because the 900 and 1800 MHz frequency bands were already allocated.

The rarer 400 and 450 MHz frequency bands are assigned in some countries, where these frequencies were previously used for first-generation systems.

GSM-900 uses 890–915 MHz to send information from the mobile station to the base station (uplink) and 935–960 MHz for the other direction (downlink), providing 124 RF channels (channel numbers 1 to 124) spaced at 200 kHz. Duplex spacing of 45 MHz is used. In some countries the GSM-900 band has been extended to cover a larger frequency range. This 'extended GSM', E-GSM, uses 880–915 MHz (uplink) and 925– 960 MHz (downlink), adding 50 channels (channel numbers 975 to 1023 and 0) to the original GSM-900 band. Time division multiplexing is used to allow eight full-rate or sixteen half-rate speech channels per radio frequency channel. There are eight radio timeslots (giving eight burst periods) grouped into what is called a TDMA frame. Half rate channels use alternate frames in the same timeslot. The channel data rate is 270.833 kbit/s, and the frame duration is 4.615 ms.

5. 7.1 GSM Advantages

GSM also pioneered a low-cost, to the network carrier, alternative to voice calls, the short t message service (SMS, also called "text messaging"), which is now supported on other mobile standards as well. Another advantage is that the standard includes one worldwide Emergency telephone number, 112. This makes it easier for international travelers to connect to emergency services without knowing the local emergency number.

5.7.2 The GSM Network

GSM provides recommendations, not requirements. The GSM specifications define the functions and interface requirements in detail but do not address the hardware. The GSM network is divided into three major systems: the switching system (SS), the base station system (BSS), and the operation and support system (OSS).

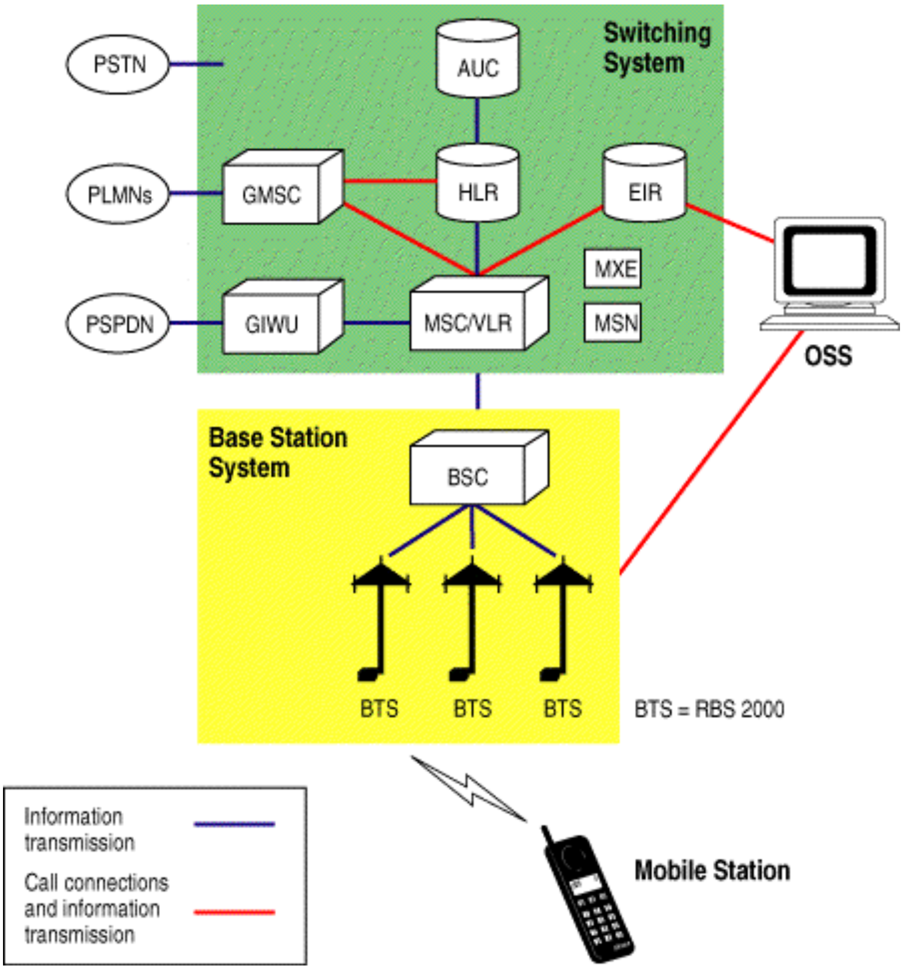

Figure 5.7.1: GSM Network

- **The Switching System:**

  The switching system (SS) is responsible for performing call processing and subscriber-related functions. The switching system includes the following functional units.

- **Home location register (HLR)**: The HLR is a database used for storage and management of subscriptions. The HLR is considered the most important database, as it stores permanent data about subscribers, including a subscriber's service profile, location information, and activity status. When an individual buys a subscription from one of the PCS operators, he or she is registered in the HLR of that operator.

- **Mobile services switching center (MSC):** The MSC performs the telephony switching functions of the system.

- **Visitor location register (VLR)**: The VLR is a database that contains temporary information about subscribers that is needed by the MSC in order to service visiting subscribers. The VLR is always integrated with the MSC. When a mobile station roams into a new MSC area, the VLR connected to that MSC will request data about the mobile station from the HLR. Later, if the mobile station makes a call, the VLR will have the information needed for call setup without having to interrogate the HLR each time.
- **Authentication center (AUC)**: A unit called the AUC provides authentication and encryption parameters that verify the user's identity and ensure the confidentiality of each call. The AUC protects network operators from different types of fraud found in today's cellular world.
- **Equipment identity register (EIR)**: The EIR is a database that contains information about the identity of mobile equipment that prevents calls from stolen, unauthorized, or defective mobile stations. The AUC and EIR are implemented as stand-alone nodes or as a combined AUC/EIR node.
- **The Base Station System (BSS):**
- All radio-related functions are performed in the BSS, which consists of base station controllers (BSCs) and the base transceiver stations (BTSs).
- **BSC**: The BSC provides all the control functions and physical links between the MSC and BTS. It is a high-capacity switch that provides functions such as handover, cell configuration data, and control of radio frequency (RF) power levels in base transceiver stations. A number of BSCs are served by an MSC.
- **BTS**: The BTS handles the radio interface to the mobile station. The BTS is the radio equipment (transceivers and antennas) needed to service each cell in the network. A group of BTSs are controlled by a BSC.

➤ **The Operation and Support System**

The operations and maintenance center (OMC) is connected to all equipment in the switching system and to the BSC. The implementation of OMC is called the operation and support system (OSS). The OSS is the functional entity from which the network operator monitors and controls the system. The purpose of OSS is to offer the customer cost-effective support for centralized, regional and local operational and maintenance activities that are required for a GSM network. An important function of OSS is to provide a network overview and support themaintenance activities of different operation and maintenance organizations.

➢ **Additional Functional Elements**

- **Message center (MXE)**: The MXE is a node that provides integrated voice, fax, and data messaging. Specifically, the MXE handles short message service, cell broadcast, voice mail, fax mail, e-mail, and notification.
- **Mobile service node (MSN)**: The MSN is the node that handles the mobile intelligent network (IN) services.
- **Gateway mobile services switching center (GMSC)**: A gateway is a node used to interconnect two networks. The gateway is often implemented in an MSC. The MSC is then referred to as the GMSC.
- **GSM inter-working unit (GIWU)**: The GIWU consists of both hardware and software that provides an interface to various networks for data communications. Through the GIWU, users can alternate between speech and data during the same call. The GIWU hardware equipment is physically located at the MSC/VLR.

5. 7.2 GSM Network Areas:

The GSM network made up of geographic areas. As shown in bellow figure, these areas include cells, location areas (LAs), MSC/VLR service areas, and public land mobile network (PLMN) areas.

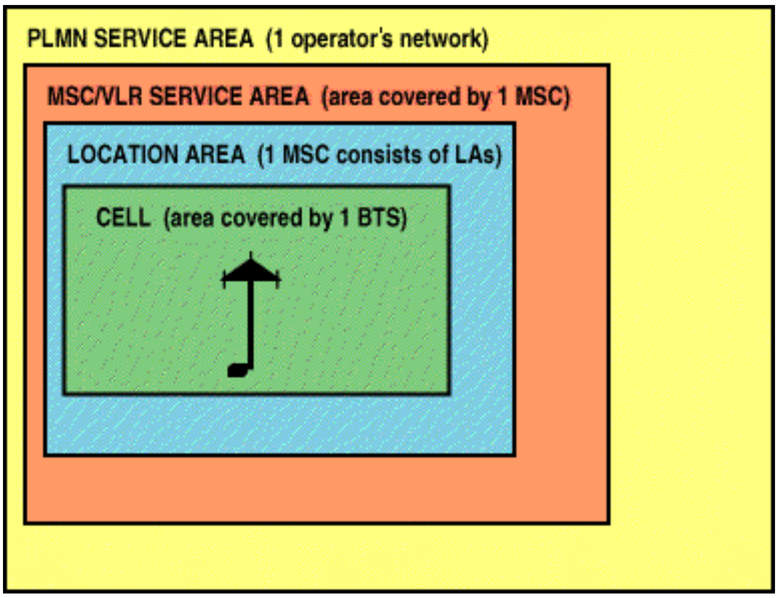

Figure 5.7.2: GSM Network Areas

- **Location Areas:**

The cell is the area given radio coverage by one base transceiver station. The GSM network identifies each cell via the cell global identity (CGI) number assigned to each cell. The location area is a group of cells. It is the area in which the subscriber is paged. Each LA is served by one or more base station controllers, yet only by a single MSC Each LA is assigned a location area identity (LAI) number.

- **MSC/VLR service areas:**

An MSC/VLR service area represents the part of the GSM network that is covered by one MSC and which is reachable, as it is registered in the VLR of the MSC. **PLMN service areas:**

The PLMN service area is an area served by one network operator.

- **GSM Specifications:**

Specifications for different personal communication services (PCS) systems vary among the different PCS networks. Listed below is a description of the specifications and characteristics for GSM.

- **Frequency band**: The frequency range specified for GSM is 1,850 to 1,990 MHz (mobile station to base station).
- **Duplex distance**: The duplex distance is 80 MHz. Duplex distance is the distance between the uplink and downlink frequencies. A channel has two frequencies, 80 MHz apart.
- **Channel separation**: The separation between adjacent carrier frequencies. In GSM, this is 200 kHz.
- **Modulation**: Modulation is the process of sending a signal by changing the characteristics of a carrier frequency. This is done in GSM via Gaussian minimum shift keying (GMSK).
- **Transmission rate**: GSM is a digital system with an over-the-air bit rate of 270 kbps.
- **Access method**: GSM utilizes the time division multiple access (TDMA) concept. TDMA is a technique in which several different calls may share the same carrier. Each call is assigned a particular time slot.
- **Speech coder**: GSM uses linear predictive coding (LPC). The purpose of LPC is to reduce the bit rate. The LPC provides parameters for a filter that mimics the vocal tract. The signal passes through this filter, leaving behind a residual signal. Speech

is encoded at 13 kbps.

- **GSM Subscriber Services:**

- **Dual-tone multifrequency (DTMF)**: DTMF is a tone signaling scheme often used for various control purposes via the telephone network, such as remote control of an answering machine. GSM supports full-originating DTMF.
- **Facsimile group III**—GSM supports CCITT Group 3 facsimile. As standard fax machines are designed to be connected to a telephone using analog signals, a special fax converter connected to the exchange is used in the GSM system. This enables a GSM– connected fax to communicate with any analog fax in the network.
- **Short message services**: A convenient facility of the GSM network is the short message service. A message consisting of a maximum of 160 alphanumeric characters can be sent to or from a mobile station. This service can be viewed as an advanced form of alphanumeric paging with a number of advantages. If the subscriber's mobile unit is powered off or has left the coverage area, the message is stored and offered back to the subscriber when the mobile is powered on or has reentered the coverage area of the network. This function ensures that the message will be received.
- **Cell broadcast**: A variation of the short message service is the cell broadcast facility. A message of a maximum of 93 characters can be broadcast to all mobile subscribers in a certain geographic area. Typical applications include traffic congestion warnings and reports on accidents.
- **Voice mail**: This service is actually an answering machine within the network, which is controlled by the subscriber. Calls can be forwarded to the subscriber's voice-mail box and the subscriber checks for messages via a personal security code.
- **Fax mail**: With this service, the subscriber can receive fax messages at any fax machine. The messages are stored in a service center from which they can be retrieved by the subscriber via a personal security code to the desired fax number.
- **Supplementary Services:** GSM supports a comprehensive set of supplementary services that can complement and support both telephony and data services.
- **Call forwarding**: This service gives the subscriber the ability to forward incoming calls to another number if the called mobile unit is not reachable, if it is busy, if there is no reply, or if call forwarding is allowed unconditionally.

- **Barring of outgoing calls**: This service makes it possible for a mobile subscriber to prevent all outgoing calls.
- **Barring of incoming calls**: This function allows the subscriber to prevent incoming calls. The following two conditions for incoming call barring exist: baring of all incoming calls and barring of incoming calls when roaming outside the home PLMN.
- **Call waiting**: This service enables the mobile subscriber to be notified of an incoming call during a conversation. The subscriber can answer, reject, or ignore the incoming call. Call waiting is applicable to all GSM telecommunications services using a circuit- switched connection.
- **Multiparty service**: The multiparty service enables a mobile subscriber to establish a multiparty conversation—that is, a simultaneous conversation between three and six subscribers. This service is only applicable to normal telephony.
- **Calling line identification presentation/restriction**: These services supply the called party with the integrated services digital network (ISDN) number of the calling party. The restriction service enables the calling party to restrict the presentation. The restriction overrides the presentation.
- **Closed user groups (CUGs)**: CUGs are generally comparable to a PBX. They are a group of subscribers who are capable of only calling themselves and certain numbers

➢ **Main AT commands:**

- "AT command set for GSM Mobile Equipment" describes the Main AT commands to communicate via a serial interface with the GSM subsystem of the phone.

- AT commands are instructions used to control a modem. AT is the abbreviation of Attention. Every command line starts with "AT" or "at". That's why modem commands are called AT commands. Many of the commands that are used to control wired dial-up modems, such as ATD (Dial), ATA (Answer), ATH (Hook control) and ATO (Return to online data state), are also supported by GSM/GPRS modems and mobile phones. Besides this common AT command set, GSM/GPRS modems and mobile phones support an AT command set that is specific to the GSM technology, which includes SMS-related commands like

AT+CMGS (Send SMS message), AT+CMSS (Send SMS message from storage), AT+CMGL (List SMS messages) and AT+CMGR (Read SMS messages).

- Note that the starting "AT" is the prefix that informs the modem about the start of a command line. It is not part of the AT command name. For example, D is the actual AT command name in ATD and +CMGS is the actual AT command name in AT+CMGS. However, some books and web sites use them interchangeably as the name of an AT command.
- Here are some of the tasks that can be done using AT commands with a GSM/GPRS modem or mobile phone:
- Get basic information about the mobile phone or GSM/GPRS modem. For example, name of manufacturer (AT+CGMI), model number (AT+CGMM), IMEI number (International Mobile Equipment Identity) (AT+CGSN) and software version (AT+CGMR).
- Get basic information about the subscriber. For example, MSISDN (AT+CNUM) and IMSI number (International Mobile Subscriber Identity) (AT+CIMI).
- Get the current status of the mobile phone or GSM/GPRS modem. For example, mobile phone activity status (AT+CPAS), mobile network registration status (AT+CREG), radio signal strength (AT+CSQ), battery charge level and battery charging status (AT+CBC).
- Establish a data connection or voice connection to a remote modem (ATD, ATA, etc).
- Send and receive fax (ATD, ATA, AT+F*).
- Send (AT+CMGS, AT+CMSS), read (AT+CMGR, AT+CMGL), write
- (AT+CMGW) or delete (AT+CMGD) SMS messages and obtain notifications of newly received SMS messages (AT+CNMI).
- Read (AT+CPBR), write (AT+CPBW) or search (AT+CPBF) phonebook entries.
- Perform security-related tasks, such as opening or closing facility locks (AT+CLCK), checking whether a facility is locked (AT+CLCK) and changing passwords (AT+CPWD).
- (Facility lock examples: SIM lock [a password must be given to the SIM card every time the mobile phone is switched on] and PH-SIM lock [a certain SIM card is associated with the mobile phone. To use other SIM cards with the mobile phone, a password must be entered.])
- Control the presentation of result codes / error messages of AT commands. For

example, you can control whether to enable certain error messages (AT+CMEE) and whether error messages should be displayed in numeric format or verbose format (AT+CMEE=1 or AT+CMEE=2).

- Get or change the configurations of the mobile phone or GSM/GPRS modem. For example, change the GSM network (AT+COPS), bearer service type (AT+CBST), radio link protocol parameters (AT+CRLP), SMS center address (AT+CSCA) and storage of SMS messages (AT+CPMS).
- Save and restore configurations of the mobile phone or GSM/GPRS modem. For example, save (AT+CSAS) and restore (AT+CRES) settings related to SMS messaging such as the SMS center address.

5.8 Global Positioning System (GPS)

The **Global Positioning System (GPS)** is the only fully functional Global Navigation Satellite System (GNSS). The GPS uses a constellation of between 24 and 32 Medium Earth Orbit satellites that transmit precise microwave signals, which enable GPS receivers to determine their location, speed, GPS was developed by the United States Department of Defense. Its official name is NAVSTAR-GPS. Although NAVSTAR- GPS is not an acronym, a few backronyms have been created for it. The GPS satellite constellation is managed by the United States Air Force 50th Space Wing.

Global Positioning System is an earth-orbiting-satellite based system that provides signals available anywhere on or above the earth, twenty-four hours a day, which can be used to determine precise time and the position of a GPS receiver in three dimensions. GPS is increasingly used as an input for Geographic Information Systems particularly for precise positioning of geospatial data and the collection of data in the field. Precise positioning is possible using GPS receivers at reference locations providing corrections and relative positioning data for remote receiver Time and frequency dissemination, based on the precise clocks on board the SVs and controlled by the monitor stations, is another, use for GPS. Astronomical observatories telecommunications facilities and laboratory standards can be set to precise time signals or controlled to accurate frequencies by special purpose GPS .Similar satellite navigation systems include the Russian GLONASS (incomplete as of

2008), the upcoming European Galileo positioning system, the proposed COMPASS navigation system of China, and IRNSS of India.

Following the shooting down of Korean Air Lines Flight 007 in 1983, President Ronald Reagan worldwide, and a useful tool for map-making, land surveying, commerce, scientific uses, and hobbies such as geocaching. GPS also provides a

Figure 5.8: GPS

5.9 L293D

The L293 and L293D are quadruple high-current half-H drivers. The L293 is designed to provide bidirectional drive currents of up to 1 A at voltages from 4.5 V to 36 V. The L293D is designed to provide bidirectional drive currents of up to 600- mA at voltages from 4.5 V to 36 V. Both devices are designed to drive inductive loads such as relays, solenoids, dc and bipolar stepping motors, as well as other high- current/high-voltage loads in positive-supply applications. All inputs are TTL compatible. Each output is a complete totem-pole drive circuit, with a Darlington transistor sink and a pseudo- Darlington source. Drivers are enabled in pairs, with drivers 1 and 2 enabled by 1,2EN and drivers 3 and 4 enabled by 3,4EN. When an enable input is high, the associated drivers are enabled, and their outputs are active and in phase with their inputs. When the enable input is low, those drivers are disabled, and their outputs are off and in the high- impedance state. With the proper data inputs, each pair of drivers forms a full-H (or bridge) reversible drive suitable for solenoid or motor applications.

Pin diagram

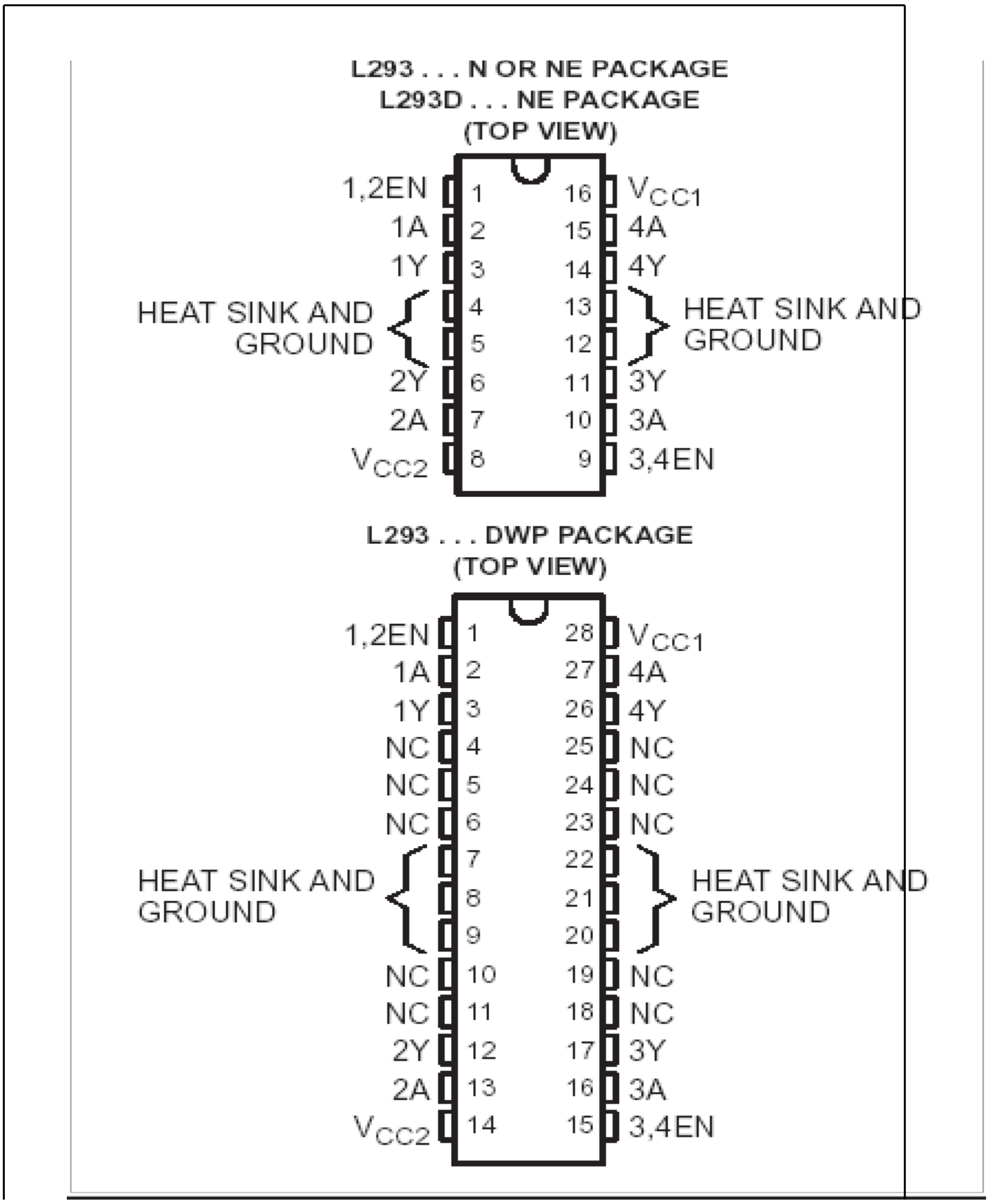

Figure 5.9: Pin diagram of L293D

5.10 DC Motor

A DC motor is designed to run on DC electric power. Two examples of pure DC designs are Michael Faraday's homopolar motor (which is uncommon), and the ball bearin motor, which is (so far) a novelty. By far the most common DC motor types are the brushed and brushless types, which use internal and external commutation respectively to create an oscillating AC current from the DC source -- so they are not purely DC machines in a strict sense.

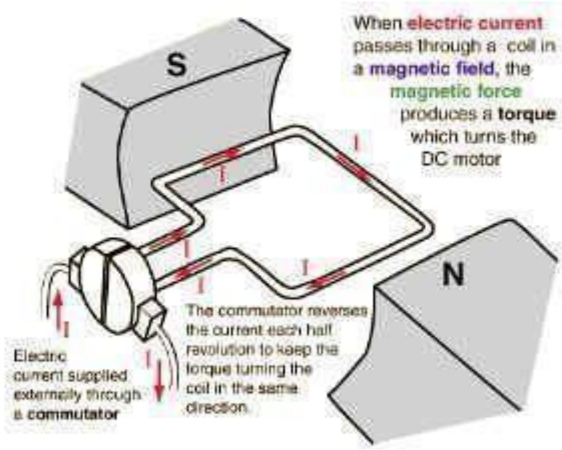

Figure 5.10: DC Motor Types of dc motors

- Brushed DC Motors
- Brushless DC motors
- Coreless DC motors

- **Brushed DC motors**

The classic DC motor design generates an oscillating current in a wound rotor with a split ring commutator, and either a wound or permanent magnet stator. A rotor consists of a coil wound around a rotor which is then powered by any type of battery. Many of the limitations of the classic commutator DC motor are due to the need for brushes to press against the commutator. This creates friction. At higher speeds, brushes have increasing difficulty in maintaining contact. Brushes may bounce off the irregularities in the commutator surface, creating sparks. This limits the maximum speed of the machine. The current density per unit area of the brushes limits the output of the motor. The imperfect electric contact also causes electrical noise. Brushes eventually wear out and require replacement, and the commutator itself is subject to wear and maintenance. The commutator assembly on a large machine is a costly element, requiring precision assembly of many parts. there are three types of dc motor 1. dc series motor 2. dc shunt motor 3. dc compound motor - these are also two type a. cumulative compound b. differcial compound.

- **Brushless DC motors:**

Some of the problems of the brushed DC motor are eliminated in the brushless design. In this motor, the mechanical "rotating switch" or commutator/brush gear assembly is replaced by an external electronic switch synchronized to the rotor's position. Brushless motors are typically 85-90% efficient, whereas DC motors with brush gear are typically 75-80% efficient. Midway between ordinary DC motors and stepper motors lies the realm of the brushless DC motor. Built in a fashion very similar to stepper motors, these often use a permanent magnet external rotor, three phases of driving coils, one or more Hall effect sensors to sense the position of the rotor, and the associated drive electronics. The coils are activated, one phase after the other, by the drive electronics as cued by the signals from the Hall effect sensors. In effect, they act as three-phase synchronous motors containing their own variable-frequency drive electronics. A specialized class of brushless DC motor controllers utilize EMF feedback through the main phase connections instead of Hall effect sensors to determine position and velocity. These motors are used extensively in electric radio-controlled vehicles. When configured with the magnets on the outside, these are referred to by modelists as out runner motors.

CHAPTER 6
SOFTWARE REQUIREMENTS

The Arduino Software (IDE) is anything but difficult to-use for fledglings, yet sufficiently adaptable for cutting edge clients to exploit too. For instructors, it's helpfully in view of the Processing programming condition, so understudies figuring out how to program in that condition will be acquainted with how the Arduino IDE functions.

Arduino is a model stage (open-source) in perspective of an easy to-use gear and programming. It includes a circuit board, which can be tweaked (suggested as a microcontroller) and a moment programming called Arduino IDE (Integrated Development Environment), which is used to make and exchange the PC code to the physical board.

The key highlights are:

1. Arduino sheets can read straightforward or propelled data signals from different sensors and change it into a yield, for instance, starting a motor, turning LED on/off, connect with the cloud and various distinctive exercises.
2. Unlike most past programmable circuit sheets, Arduino does not require an extra piece of gear (considered a product build) with a particular ultimate objective to stack another code onto the board. You can simply use a USB interface.
3. Additionally, the Arduino IDE uses a streamlined interpretation of C++, making it less requesting to make sense of how to program.
4. Finally, Arduino gives a standard edge factor that breaks the components of the scaled down scale controller into a more open package.

6.1 Operation Demo:

STEP 1: Install The Arduino Software (IDE)

Download the most recent variant from this page: http://arduino.cc/en/Main/Software

Next, continue with the establishment and please permit the driver establishment process

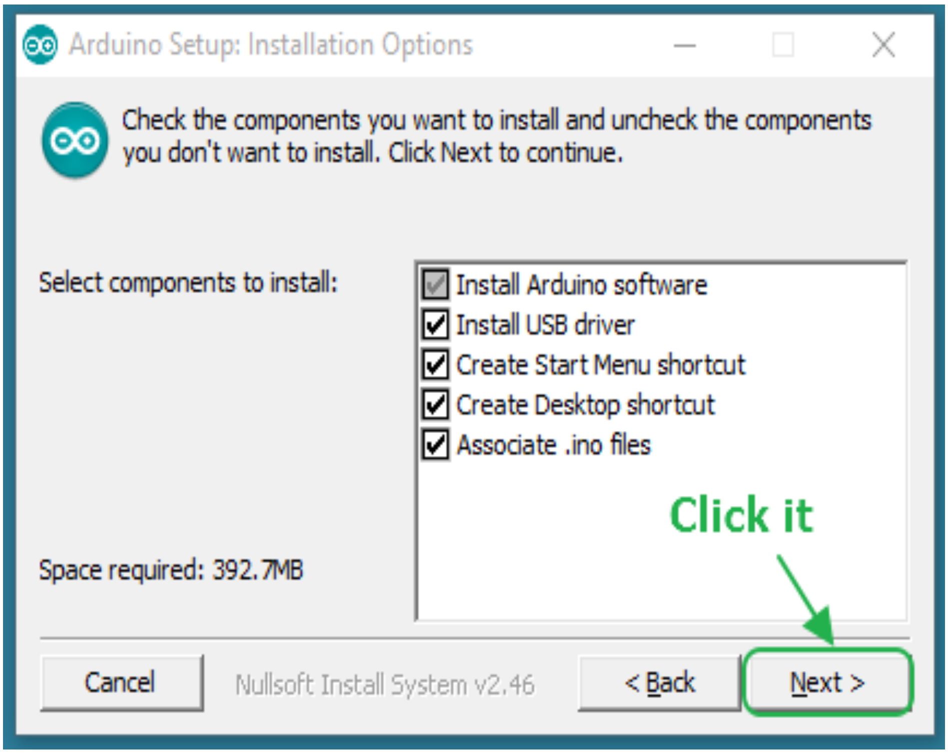

Figure 6.1.1: Arduino setup-installation options

Pick the parts to introduce and click "next" catch

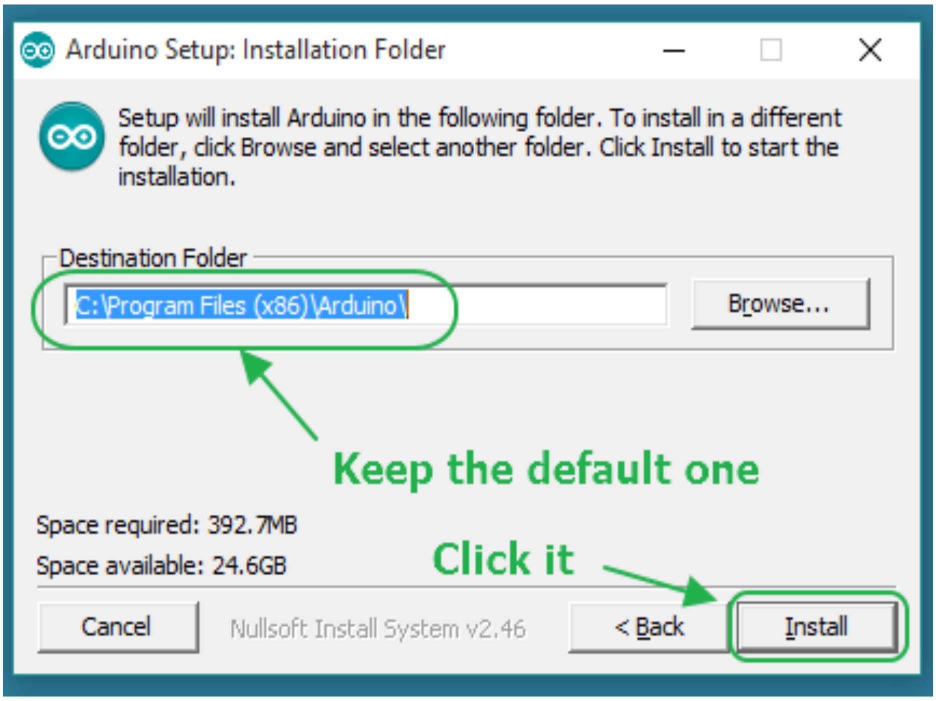

Figure 6.1.2: Arduino setup installation folder

Pick the establishment index.

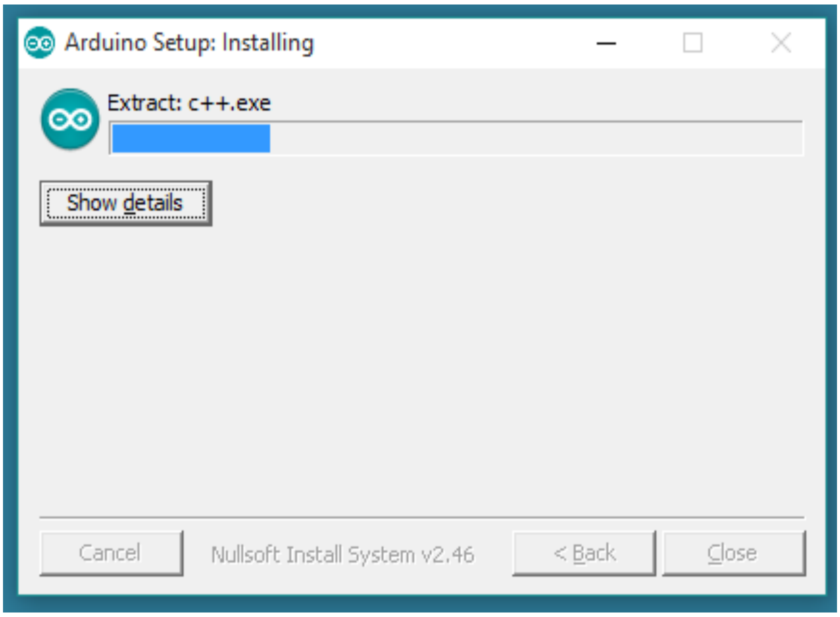

Figure 6.1.3: Arduino setup-installing

The procedure will separate and introduce all the expected documents to execute legitimately the Arduino Software (IDE)

STEP 2: Get An UNO R3 And USB Cable

In this instructional exercise, you're utilizing an Uno R3. You additionally require a standard USB link (A fitting to B plug): the kind you would associate with a USB printer, for instance.

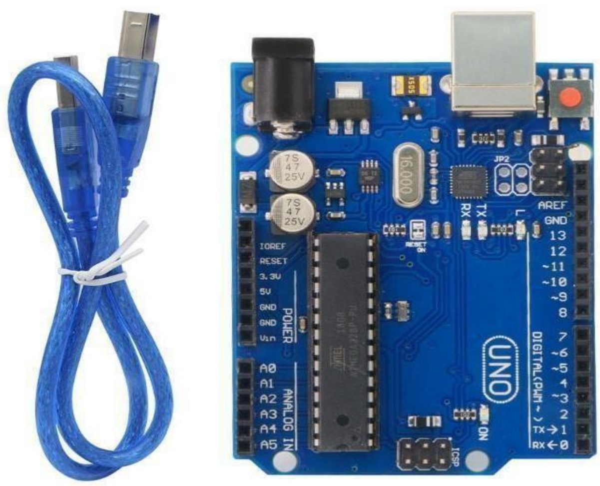

Figure 6.1.4: USB cable UNO R3 board

STEP 3: Connect The Board

The USB association with the PC is important to program The USB association with the PC is important to program the board and not simply to control it up. The Uno and

Mega consequently draw control from either the USB or an outside power supply. Associate the board to your PC utilizing the USB link.

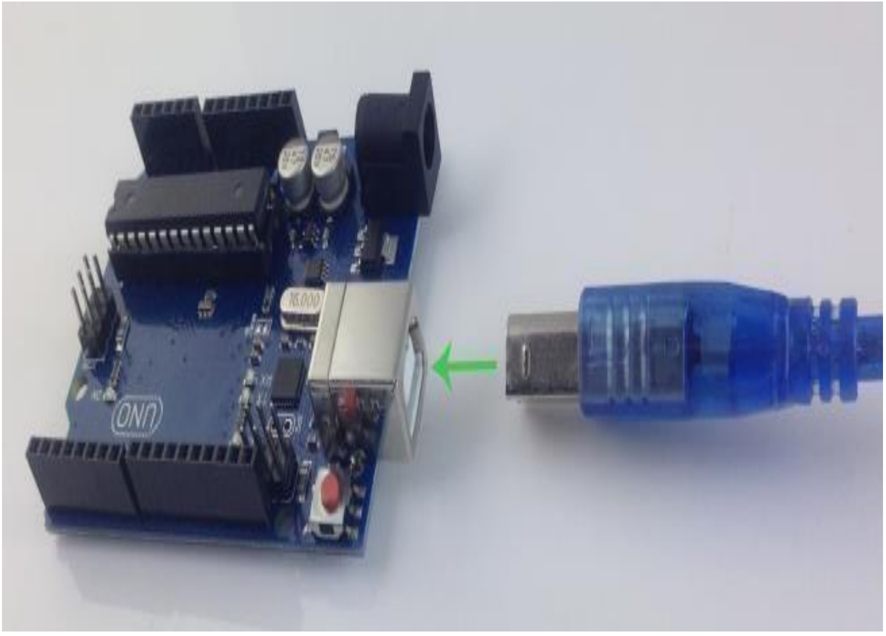

Figure 6.1.5: Represents how to connect USB cable with UNO board

STEP 4: Open Lesson 1: LED Blink

Open the LED blink example sketch: CD > For Arduino>Demo Code>Lesson1-LED_bink>led blink.

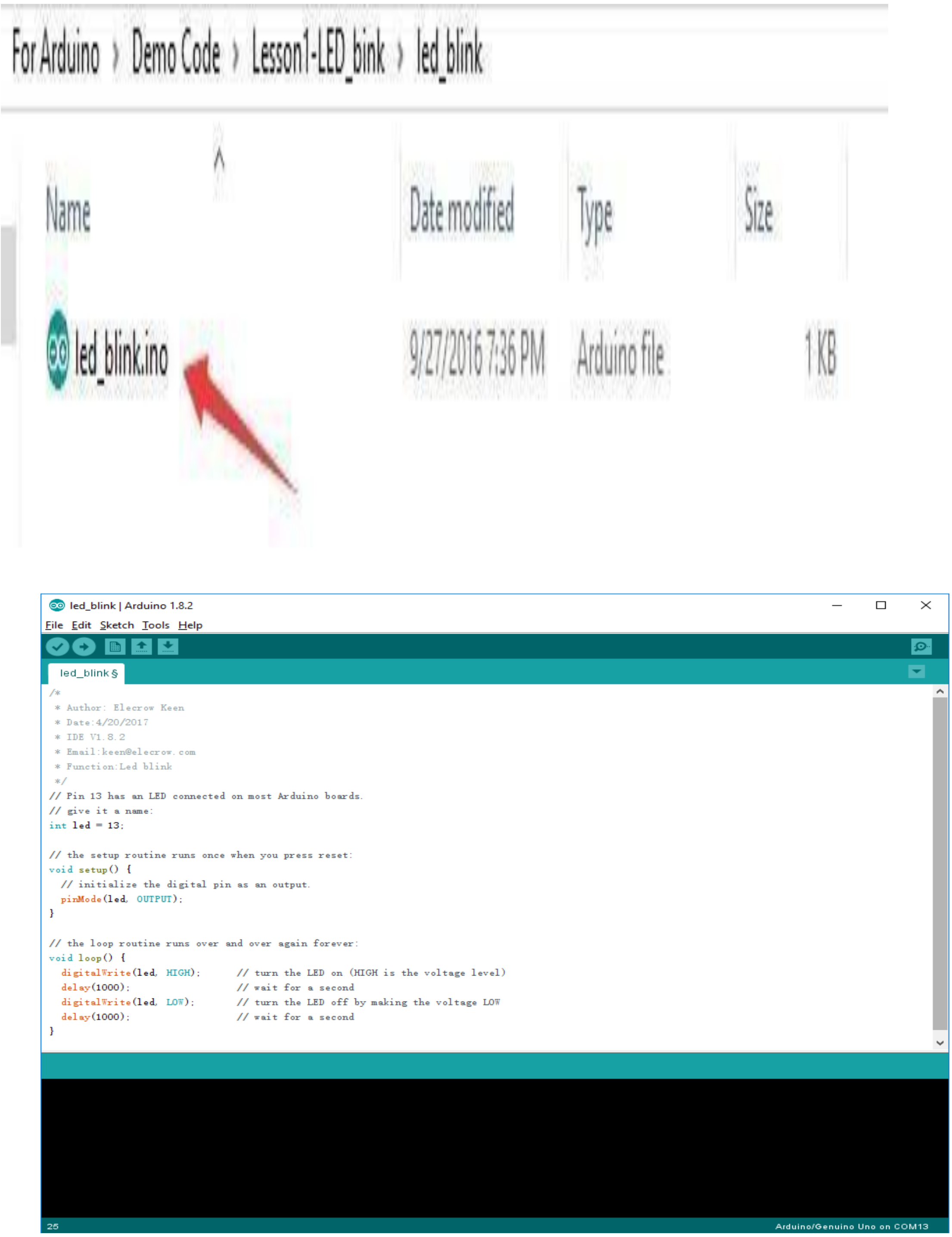

Figure 6.1.6: LED valve and program to blink LED

STEP 5: Select Your Board

You'll need to select the entry in the Tools > Board menu that corresponds to your Arduino board.

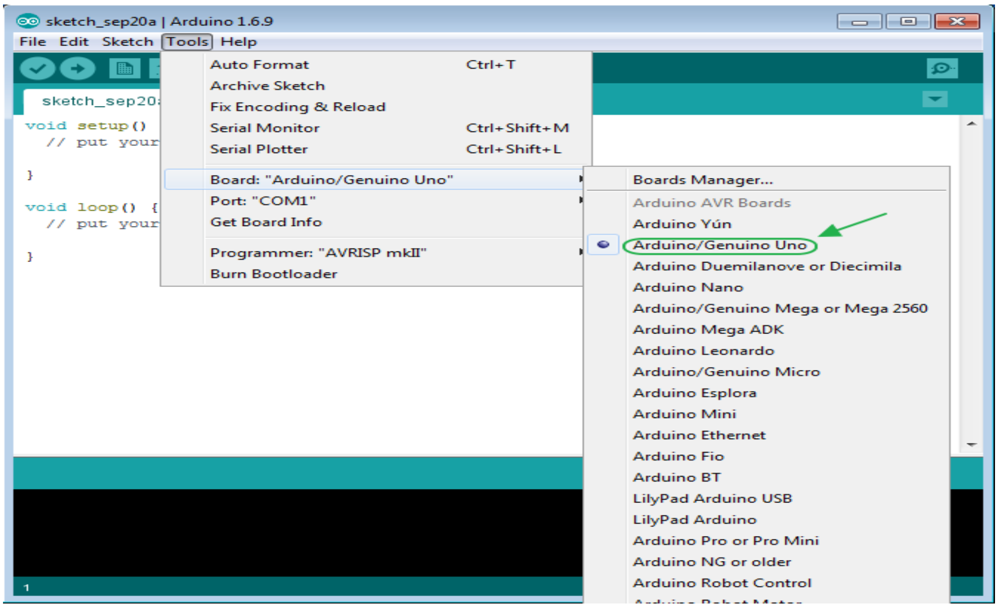

Figure 6.1.7: path to select board

Selecting an Arduino/Genuine Uno

STEP 6: Select Your Serial Port

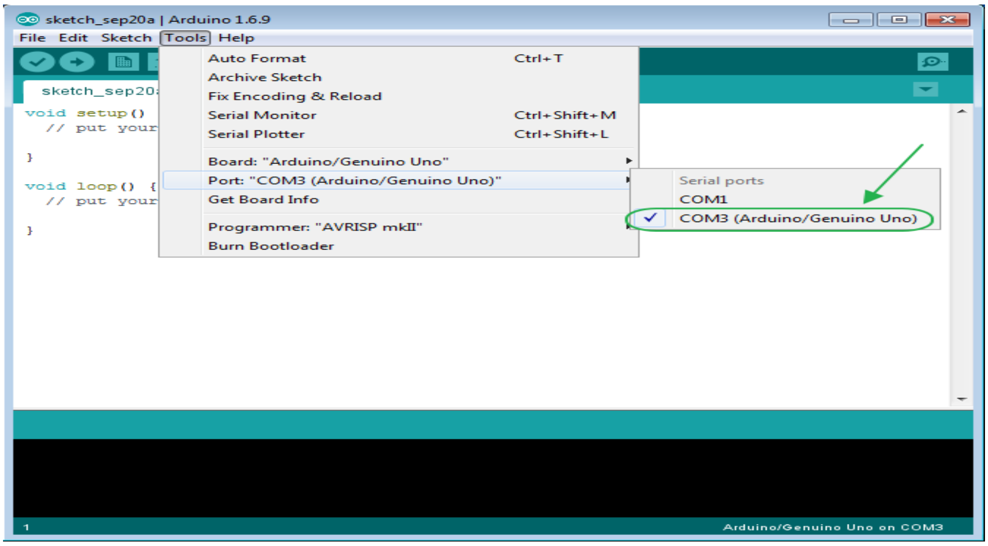

Figure 6.1.8: path to select serial port

Select the serial gadget of the board from the Tools | Serial Port menu. This is probably going to be COM3or higher (COM1andCOM2are generally held for equipment serial ports). To discover, you can detach your board and re-open the menu; the passage that vanishes ought to be the Arduino board. Reconnect the board and select that serial port

STEP 7: Upload The Program

Presently, essentially tap the "Transfer" catch in the earth. Hold up a couple of moments - you should see the RX and TX leds on the board blazing. On the off chance that the transfer is fruitful, the message "Done transferring." will show up in the status bar.

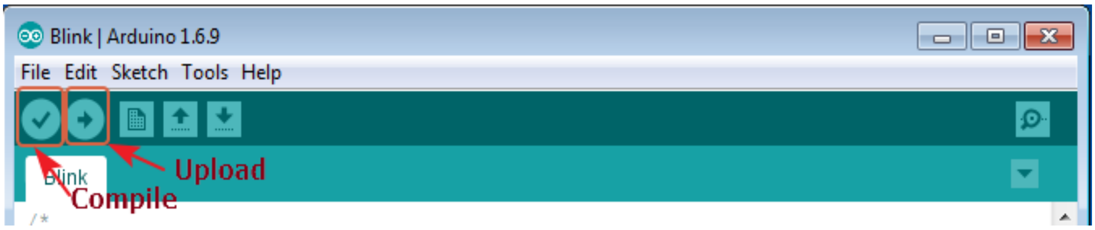

Figure 6.1.9: path to upload

STEP 8: Result

A few seconds after the upload finishes, you should see the pin 13 (L) LED on the board start to blink (in orange). If it does, congratulations! You've gotten Arduino up-and-running.

6.2 Arduino Interface Introduction

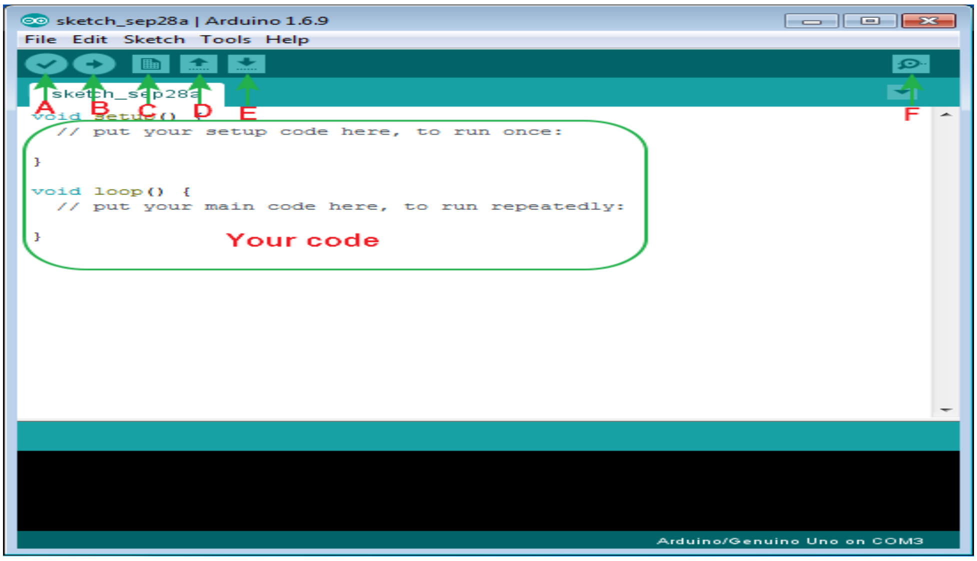

Figure 6.2: Tools in Arduino interface software

A ->Compile
B ->Upload

C ->New
D ->Open
E ->Save
F ->Serial monitor

6.3 Arduino UNO R3 Hardware Introduction

Figure 6.3: UNO with hardware

6.4 How To Add Library Files

Step 1:

Add library file: Sketch>Include Library>Add.ZIP Library

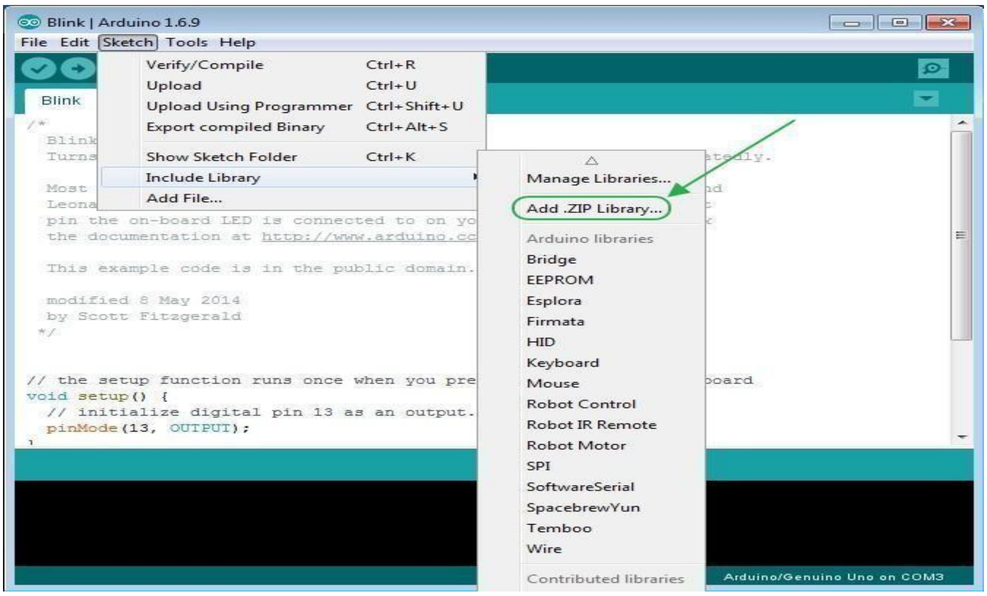

Figure 6.4.1: path to add library files

STEP 2:

Select your library file compression package on the demo code file, as follows:

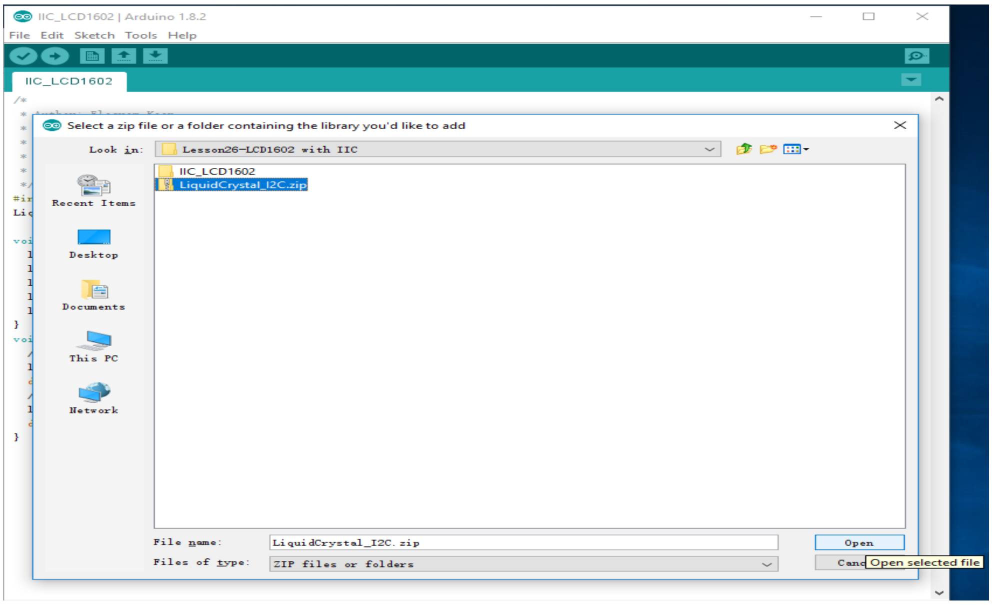

Figure 6.4.2: selecting the file to Arduino

CHAPTER 7

RESULTS & DISCUSSIONS

7.1 Results

The system is designed to enhance road safety by preventing drunk driving and facilitating emergency response in the event of an accident or fire. To accomplish this, the system includes an automated mechanism that disables vehicle operation when the driver is detected to be under the influence of alcohol. This system also stores the whole data into the IoT platform which is Thingspeak. In the case of a collision, the system automatically sends the vehicle's GPS location to a designated emergency contact. The Internet of Things (IoT) operates by interconnecting physical devices with the internet to collect, share, and act upon data in real-time. These devices are embedded with sensors that monitor specific parameters such as temperature, fire, gas, angle. In an accident monitoring system, sensors like accelerometers, GPS modules, and impact detectors play a crucial role in identifying abnormal situations or collisions. Overall, the system is designed to improve traffic safety by preventing impaired driving and enabling rapid emergency intervention in critical situations and can analyse about the accident in Thingspeak platform.

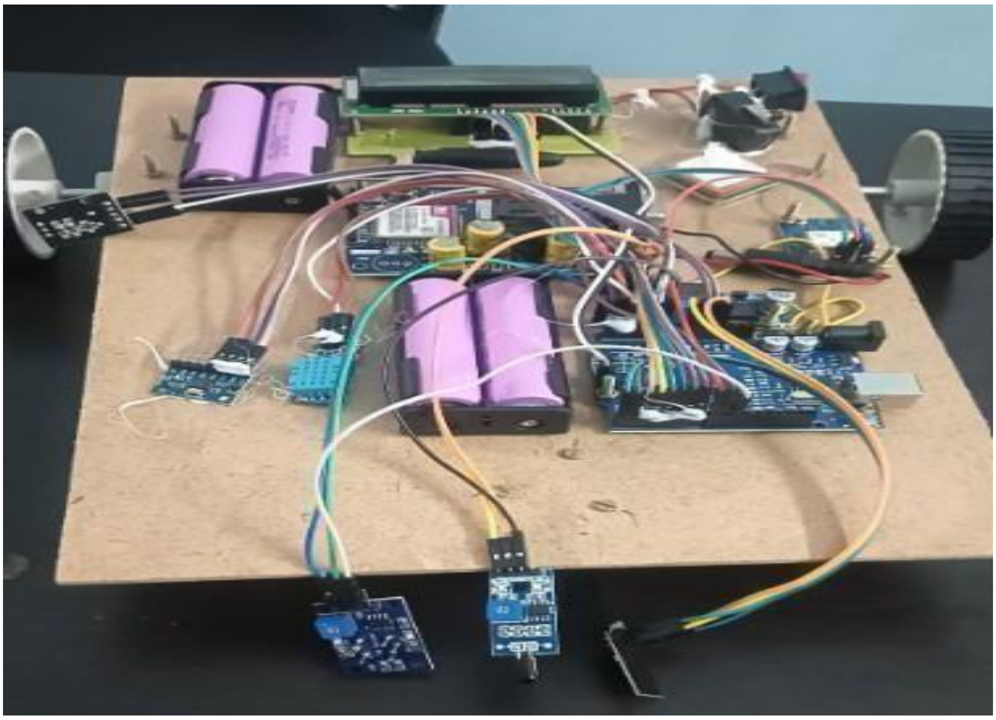

Figure 7.1.1: Hardware of Real-Time Accident Monitoring System

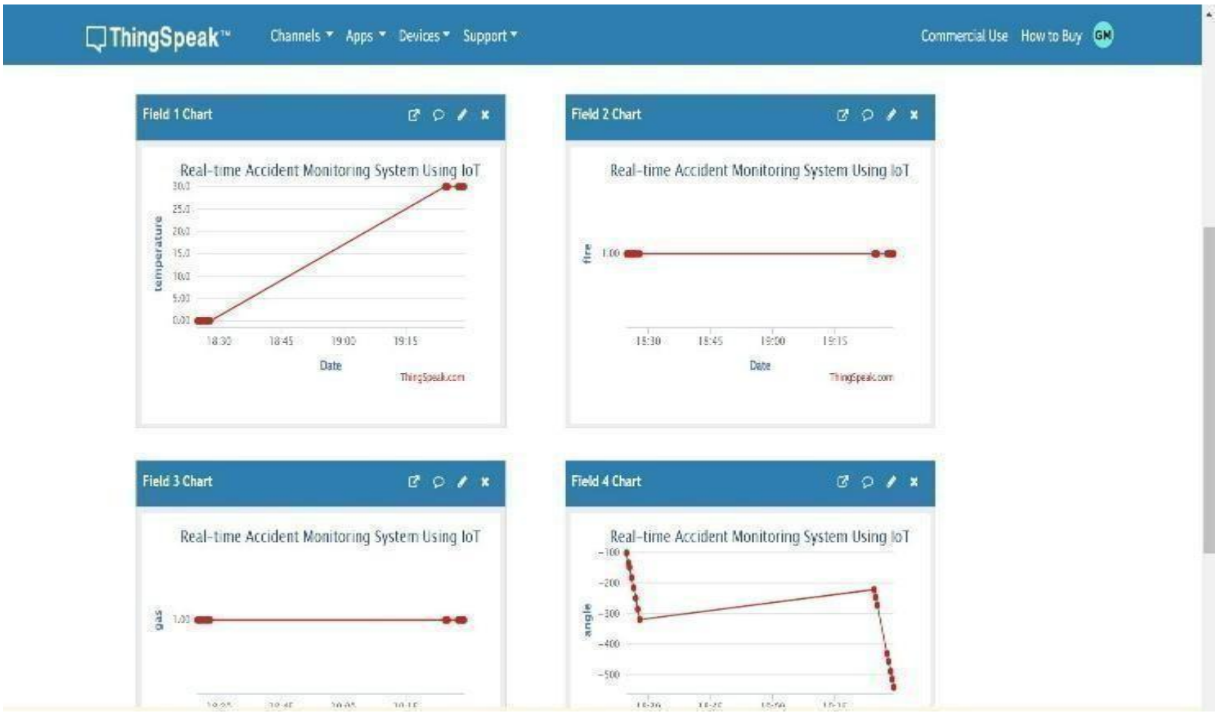

Figure 7.1.2: Thingspeak Output

The image displays a dashboard from ThingSpeak showing real-time data visualizations for an IoT-based accident monitoring system. There are four separate field charts, each representing different sensor readings: temperature, fire detection, gas level, and angle. These parameters are crucial for detecting potential accidents or hazardous events. The temperature chart shows a significant increase, indicating a possible rise in environmental heat. The fire and gas readings appear constant, while the angle chart shows fluctuations that could suggest a tilt or impact—useful for detecting vehicle accidents. Together, these graphs demonstrate how IoT can be utilized for continuous monitoring and quickresponse in accident scenarios.

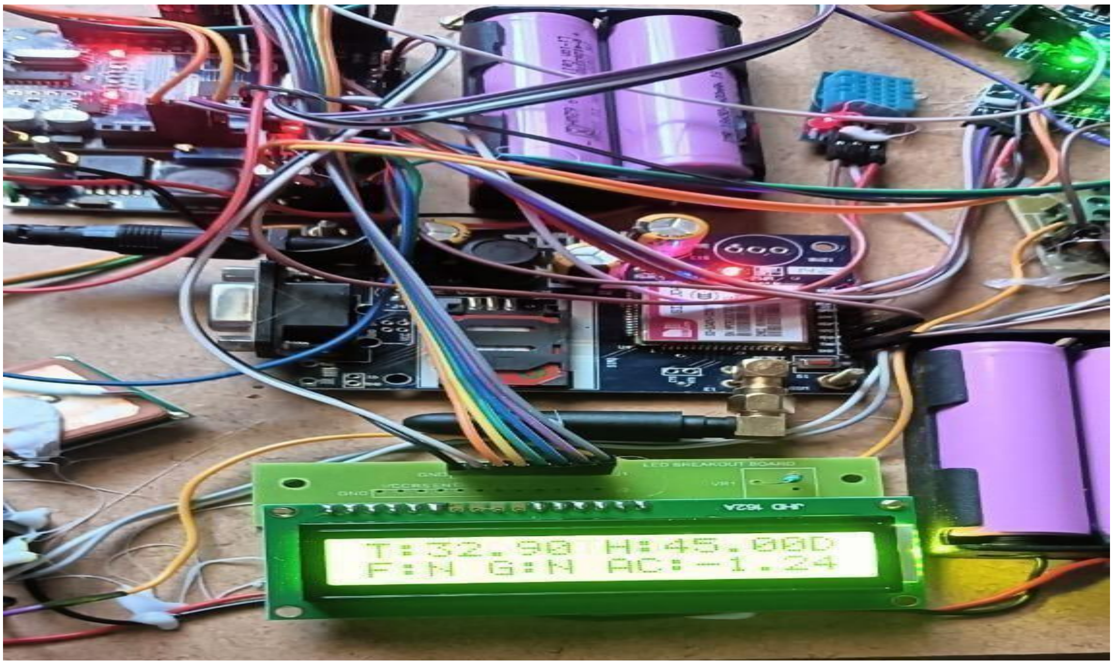

Figure 7.1.3: LCD Values

This shows a prototype of an IoT-based accident monitoring system built using various electronic components. At the center, there's an Arduino microcontroller connected to a GSM module, a GPS module, sensors (such as a temperature and humidity sensor), and a display. The LCD screen is actively displaying real-time data including temperature (T: 32.90°C), humidity (H: 45.00%), fire detection (F: N), gas detection (G: N), and acceleration (AC: 1.24), indicating the system is functioning and collecting live environmental and motion data. Powered by rechargeable batteries, this compact setup is designed for use in smart safety systems—especially for vehicle accident detection and alerting—by collecting critical parameters and transmitting them remotely via GSM.

Figure 7.1.4: LCD Output

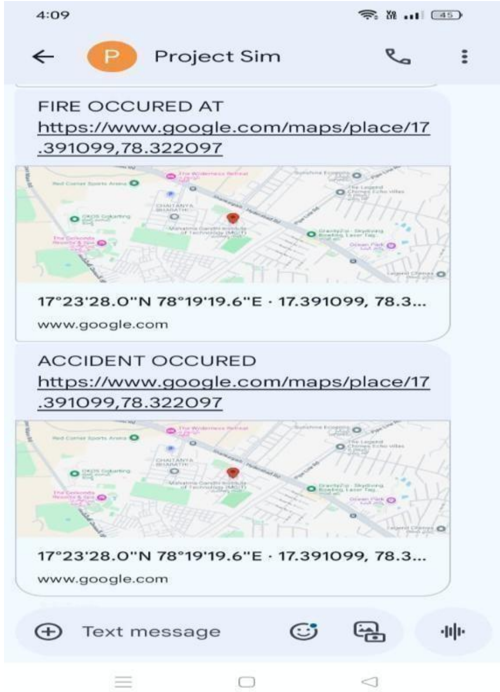

Figure 7.1.5: SMS

The image shows a mobile text message alert system from an IoT-based accident and fire monitoring project. The messages, sent from a SIM module ,contain real-time notifications of a fire and an accident, each with an embedded Google Maps link and corresponding GPS coordinates (17°23'28.0"N, 78°19'19.6"E). This system demonstrates how emergencies such as fires or vehicle accidents are detected and immediately communicated with exact location details, enabling quick response and efficient emergency handling. It's a key feature in smart safety systems for real-time incident tracking.

7.2 Discussions

Enhancing road safety by actively preventing drunk driving through the use of an alcohol detection sensor. When alcohol is detected in the driver’s breath, the vehicle’s ignition is automatically disabled, reducing the risk of impaired driving-related accidents. This preventive feature ensures that individuals under the influence cannot operate the vehicle, thereby promoting responsible driving habits and reducing the likelihood of collisions.

In the event of a collision, the system is programmed to automatically send the vehicle's GPS coordinates to a designated emergency contact. This enables quick identification of the accident site, allowing emergency services to respond faster and potentially save lives. Additionally, if a fire is detected within the vehicle, the system alerts relevant fire departments by sending the exact location, allowing them to respond promptly and minimize damage.

A key component of the system is its integration with the IoT platform, Thingspeak. This platform is used to store, visualize, and analyze all system data in real time. Events such as alcohol detection, crash alerts, fire incidents, and GPS location updates are continuously uploaded to Thingspeak. This allows users and authorities to monitor vehicle safety status remotely and maintain a historical log of incidents. The graphical data representation on the platform helps in analyzing trends, identifying high-risk zones, and planning preventive strategies. This adds an important layer of intelligence and transparency to the system.

The automation and real-time capabilities of the system make it highly efficient with minimal human intervention. However, it is important to note that the system relies on accurate sensors and consistent network connectivity. In remote areas or environments with signal issues, data transmission may be delayed or disrupted, which could affect the system's performance.

7.3 Applications

- Prevents drunk driving by stopping the vehicle if the driver is under the influence of alcohol
- Improves road safety by reducing the chances of accidents caused by impaired driving
- Sends emergency alerts with GPS location to a contact person when a crash happens
- Helps emergency teams find the accident spot quickly and provide fast help
- Detects vehicle fires and sends the vehicle's location to fire services for quick action
- Stores all data (like alcohol level, accidents, fire, GPS info) on the Thingspeak IoT platform
- Allows accident analysis on Thingspeak, which can help understand causes and patterns of accidents

CHAPTER 8

CONCLUSION

8.1 Conclusion

The proposed system significantly enhances road safety by integrating automatic vehicle shutdown upon detecting driver intoxication, real-time GPS location transmission during collisions, and fire detection alerts. The Internet of Things (IoT) operates by interconnecting physical devices with the internet to collect, share, and act upon data in real-time. These features collectively aim to reduce drunk driving incidents, improve emergency response times, and minimize casualties in accidents. By leveraging advanced sensors and communication technologies, this system provides a proactive approach to accident prevention and rapid emergency assistance. Additionally, the system’s automation ensures that responses are immediate and do not rely on human intervention, thereby increasing efficiency in critical situations. Overall, it contributes to a safer driving environment by preventing accidents and ensuring timely help in emergencies.

8.2 Future Scope

Future enhancements can include integrating artificial intelligence for more accurate intoxication detection, real-time health monitoring of drivers, and automated emergency braking in critical situations. Additionally, the system could incorporate machine learning to analyze driving patterns and provide early warnings for potential drowsiness or impairment. Enhancements such as voice-assisted alerts, cloud-based data storage for accident analysis, and integration with vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication can further enhance safety and response efficiency. Moreover, expanding connectivity with smart city infrastructure and emergency services can enable seamless coordination with law enforcement and medical teams, ensuring quicker assistance. The incorporation of blockchain technology for secure data sharing in accident scenarios could also be explored, adding reliability and transparency to emergency response mechanisms.

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APPENDIX

SOURCE CODE

```
#include <SoftwareSerial.h>
#include <LiquidCrystal.h>
#include <TinyGPS++.h>
#include <Wire.h>
TinyGPSPlus gps;
#include <DFRobot_DHT11.h>
DFRobot_DHT11 DHT;
#define DHT11_PIN A3
LiquidCrystal lcd(2, 3, 4, 5, 6, 7);
const char* ssid = "project";
const char* password = "project1234";
String apiKey = "W3WP6C8BY0ECXM2K";
const int fire = A0;
const int R_outA = A2;
const int R_outB = A1;
const int gas = 10;
const int motor = 11;
SoftwareSerial gsm_gps(8,9);
int x,y,z;
int counter = 0;
int aState;
int aLastState;
float lat = 17.3911;
float logt = 78.3221;//MGIT
float t = 34.0;
float h = 45.0;
String mobilenumber = "AT+CMGS=\"+917013747561\"\r";
String message= "WATER EVACUATION";
/******/
```

```
const int MPU = 0x68; // MPU6050 I2C address
float AccX, AccY, AccZ;
float GyroX, GyroY, GyroZ;
float accAngleX, accAngleY, gyroAngleX, gyroAngleY, gyroAngleZ;
float roll, pitch, yaw;
float AccErrorX, AccErrorY, GyroErrorX, GyroErrorY, GyroErrorZ;
float elapsedTime, currentTime, previousTime;
int c = 0;
/******/
void setup() {
  //dht.begin();
  Serial.begin(115200);
  gsm_gps.begin(9600);
  pinMode(fire,INPUT);
  pinMode(R_outA,INPUT);
  pinMode(R_outB,INPUT);
  pinMode(gas,INPUT);
  pinMode(motor,OUTPUT);
  digitalWrite(motor,HIGH);
  lcd.begin(16, 2);
  lcd.setCursor(0,0);
  lcd.print("VEHICLE SPEAD ");
  lcd.setCursor(5,1);
  lcd.print("GSM GPS  ");
  delay(1000);
  gsm_gps.println("AT");
  delay(1000);
  gsm_gps.println("AT+CNMI=2,2,0,0,0");
  delay(500);
  gsm_gps.println("AT+CMGF=1");
  send_sms(mobilenumber,"TEST SMS");
  lcd.clear();
  /****/
```

```
  Wire.begin();                     // Initialize comunication
   Wire.beginTransmission(MPU);             // Start communication  with MPU6050 //
MPU=0x68
  Wire.write(0x6B);                  // Talk to the register 6B
  Wire.write(0x00);                  // Make reset - place a 0 into the 6B register
  Wire.endTransmission(true);        //end the transmission
  /*
  // Configure Accelerometer Sensitivity - Full Scale Range (default +/- 2g)
  Wire.beginTransmission(MPU);
  Wire.write(0x1C);                  //Talk to the ACCEL_CONFIG register (1C hex)
  Wire.write(0x10);                  //Set the register bits as 00010000 (+/- 8g full scale range)
  Wire.endTransmission(true);
  // Configure Gyro Sensitivity - Full Scale Range (default +/- 250deg/s)
  Wire.beginTransmission(MPU);
  Wire.write(0x1B);                   // Talk to the GYRO_CONFIG register (1B hex)
  Wire.write(0x10);                  // Set the register bits as 00010000 (1000deg/s full scale)
  Wire.endTransmission(true);
  delay(20);
  */
  // Call this function if you need to get the IMU error values for your module
  calculate_IMU_error();
  delay(20);
  /****/
  aLastState = digitalRead(R_outA);
  connectToWiFi();
}
void loop() {
  DHT.read(DHT11_PIN);
  DHT11a();
  mpu6050();
  x = digitalRead(fire);
  y = digitalRead(gas);
```

```
 rotary();
 lcd.setCursor(14,0);
 lcd.print(counter);
 if(x == LOW)
 {
 lcd.setCursor(0,1);
 lcd.print("F:Y");
 get_gps();
 delay(2000);
 lcd.clear();
 lcd.setCursor(0,1);
 lcd.print("SMS SENDING 1");
 send_sms(mobilenumber,"FIRE OCCURED AT");
 delay(1000);
 lcd.clear();
 }
else
{
 lcd.setCursor(0,1);
 lcd.print("F:N");
}
if(y == LOW)
{
 lcd.setCursor(3,1);
 lcd.print(" G:Y");
 send_sms(mobilenumber,"GAS DETECTED");
 digitalWrite(motor,LOW);
 delay(1000);
 lcd.clear();
}
else
{
```

```
  lcd.setCursor(3,1);
  lcd.print(" G:N");
  digitalWrite(motor,HIGH);
 }
lcd.setCursor(7,1);
lcd.print("  AC:");
lcd.print(GyroY);
/*if(GyroY                    >                    10)
send_sms(mobilenumber,"ACCIDENT OCCURED");

delay(1000);
lcd.clear();
lcd.clear();
lcd.setCursor(0,0);
lcd.print("ACCIDENT OCCURED");*/
  z = z + 1;
  Serial.println(z);
  if(z> 20)
  {
  z = 0;
  uploadToThingSpeak(DHT.temperature,x,y,pitch,pitch);
  }
}
void send_sms(String mnumber,String str)
{
    gsm_gps.println("AT");
    delay(500);
    gsm_gps.println("AT+CMGF=1");
    delay(500);
    //gsm.println("AT+CMGS=\"+916300102942\"\r");
    gsm_gps.println(mnumber);
    delay(1000);
```

```
                 //Serial.println("TROUBLE                        AT
https://www.google.com/maps/place/15.4340,80.0467");
   gsm_gps.print(str);
   gsm_gps.print(" https://www.google.com/maps/place/");
   gsm_gps.print(lat,6);
   gsm_gps.print(",");
   gsm_gps.println(logt,6);
   delay(2000);
   gsm_gps.println((char)26); // ASCII code of CTRL+Z
   delay(1000);
   gsm_gps.println("AT");
}
void get_gps()
{
  lcd.clear();
  for(int i=0;i<80;i++)
  {
while (gsm_gps.available() > 0){
   gps.encode(gsm_gps.read());
   if(gps.location.isUpdated()){
    /* Serial.print("Latitude= ");
     Serial.print(gps.location.lat(), 6);
     Serial.print(" Longitude= ");
     Serial.println(gps.location.lng(), 6);*/
     lcd.setCursor(0,0);
     lcd.print("LAT:");
     lcd.print(gps.location.lat(), 6);
     lcd.setCursor(0,1);
     lcd.print("LOG:");
     lcd.print(gps.location.lng(), 6);
     lat = gps.location.lat();
     logt = gps.location.lng();
```

```
      delay(1000);
    }
  }
  delay(80);
}
      lcd.setCursor(0,0);
      lcd.print("LAT:");
      lcd.print(lat, 6);
      lcd.setCursor(0,1);
      lcd.print("LOG:");
      lcd.print(logt, 6);
}
void DHT11a()
{
//int chk = DHT.read11(DHT11_PIN);
 /* Serial.print("Temperature = ");
  Serial.println(DHT.temperature);
  Serial.print("Humidity = ");
  Serial.println(DHT.humidity);*/

lcd.setCursor(0,0);
lcd.print("T:");
lcd.print(DHT.temperature);
lcd.print(" H:");
lcd.print(DHT.humidity);
}
void mpu6050()
{
  Wire.beginTransmission(MPU);
  Wire.write(0x3B); // Start with register 0x3B (ACCEL_XOUT_H)
  Wire.endTransmission(false);
  Wire.requestFrom(MPU, 6, true); // Read 6 registers total, each axis value is stored in
```

```
2 registers
  //For a range of +-2g, we need to divide the raw values by 16384, according to the datasheet
  AccX = (Wire.read() << 8 | Wire.read()) / 16384.0; // X-axis value
  AccY = (Wire.read() << 8 | Wire.read()) / 16384.0; // Y-axis value
  AccZ = (Wire.read() << 8 | Wire.read()) / 16384.0; // Z-axis value
  // Calculating Roll and Pitch from the accelerometer data
  accAngleX = (atan(AccY / sqrt(pow(AccX, 2) + pow(AccZ, 2))) * 180 / PI) - 0.58; // AccErrorX ~(0.58) See the calculate_IMU_error()custom function for more details
  accAngleY = (atan(-1 * AccX / sqrt(pow(AccY, 2) + pow(AccZ, 2))) * 180 / PI) + 1.58; // AccErrorY ~(-1.58)
  // === Read gyroscope data === //
  previousTime = currentTime;        // Previous time is stored before the actual time read
  currentTime = millis();            // Current time actual time read
  elapsedTime = (currentTime - previousTime) / 1000; // Divide by 1000 to get seconds
  Wire.beginTransmission(MPU);
  Wire.write(0x43); // Gyro data first register address 0x43
  Wire.endTransmission(false);
  Wire.requestFrom(MPU, 6, true); // Read 4 registers total, each axis value is stored in 2 registers
  GyroX = (Wire.read() << 8 | Wire.read()) / 131.0; // For a 250deg/s range we have to divide first the raw value by 131.0, according to the datasheet
  GyroY = (Wire.read() << 8 | Wire.read()) / 131.0;
  GyroZ = (Wire.read() << 8 | Wire.read()) / 131.0;
  // Correct the outputs with the calculated error values
  GyroX = GyroX + 0.56; // GyroErrorX ~(-0.56)
  GyroY = GyroY - 2; // GyroErrorY ~(2)
  GyroZ = GyroZ + 0.79; // GyroErrorZ ~ (-0.8)
  // Currently the raw values are in degrees per seconds, deg/s, so we need to multiply by sendonds (s) to get the angle in degrees
  gyroAngleX = gyroAngleX + GyroX * elapsedTime; // deg/s * s = deg
  gyroAngleY = gyroAngleY + GyroY * elapsedTime;
```

```
  yaw =  yaw + GyroZ * elapsedTime;
  // Complementary filter - combine acceleromter and gyro angle values
  roll = 0.96 * gyroAngleX + 0.04 * accAngleX;
  pitch = 0.96 * gyroAngleY + 0.04 * accAngleY;

  // Print the values on the serial monitor
  /*Serial.print(roll);
  Serial.print("/");
  Serial.print(pitch);
  Serial.print("/");
  Serial.println(yaw);*/
  /*Serial.print("X=");
  Serial.print(GyroX);
  Serial.print("Y=");
  Serial.print(GyroX);
  Serial.print("Z=");
  Serial.println(GyroX);*/
}
void calculate_IMU_error() {
  // We can call this funtion in the setup section to calculate the accelerometer and gyro data error. From here we will get the error values used in the above equations printed on the Serial Monitor.
  // Note that we should place the IMU flat in order to get the proper values, so that we then can the correct values
  // Read accelerometer values 200 times
  while (c < 200) {
  Wire.beginTransmission(MPU);
  Wire.write(0x3B);
  Wire.endTransmission(false);
  Wire.requestFrom(MPU, 6, true);
    AccX = (Wire.read() << 8 | Wire.read()) / 16384.0 ;
    AccY = (Wire.read() << 8 | Wire.read()) / 16384.0 ;
```

```
    AccZ = (Wire.read() << 8 | Wire.read()) / 16384.0 ;
    // Sum all readings
    AccErrorX = AccErrorX + ((atan((AccY) / sqrt(pow((AccX), 2) + pow((AccZ), 2)))
* 180 / PI));
    AccErrorY = AccErrorY + ((atan(-1 * (AccX) / sqrt(pow((AccY), 2) + pow((AccZ),
2))) * 180 / PI));
    c++;
  }
  //Divide the sum by 200 to get the error value
  AccErrorX = AccErrorX / 200;
  AccErrorY = AccErrorY / 200;
  c = 0;
  // Read gyro values 200 times
  while (c < 200) {
  Wire.beginTransmission(MPU);
  Wire.write(0x43);
  Wire.endTransmission(false);
  Wire.requestFrom(MPU, 6, true);
    GyroX = Wire.read() << 8 | Wire.read();
    GyroY = Wire.read() << 8 | Wire.read();
    GyroZ = Wire.read() << 8 | Wire.read();
    // Sum all readings
    GyroErrorX = GyroErrorX + (GyroX / 131.0);
    GyroErrorY = GyroErrorY + (GyroY / 131.0);
    GyroErrorZ = GyroErrorZ + (GyroZ / 131.0);
    c++;
  }
  //Divide the sum by 200 to get the error value
  GyroErrorX = GyroErrorX / 200;
  GyroErrorY = GyroErrorY / 200;
  GyroErrorZ = GyroErrorZ / 200;
  // Print the error values on the Serial Monitor
```

```
  /*Serial.print("AccErrorX: ");
  Serial.println(AccErrorX);
  Serial.print("AccErrorY: ");
  Serial.println(AccErrorY);
  Serial.print("GyroErrorX: ");
  Serial.println(GyroErrorX);
  Serial.print("GyroErrorY: ");
  Serial.println(GyroErrorY);
  Serial.print("GyroErrorZ: ");
  Serial.println(GyroErrorZ);*/
}
void rotary()
{
  aState = digitalRead(R_outA); // Reads the "current" state of the outputA
   // If the previous and the current state of the outputA are different, that means a Pulse
has occured
   if (aState != aLastState){
     // If the outputB state is different to the outputA state, that means the encoder is
rotating clockwise
     if (digitalRead(R_outB) != aState) {
       counter ++;
     } else {
       counter --;
     }
     //Serial.print("Position: ");
     //Serial.println(counter);
   }
   aLastState = aState;
}

void connectToWiFi() {
  //Serial.println("Connecting to Wi-Fi...");
```

```
  sendCommand("AT", 1500);  // Test communication
  sendCommand("AT+CWMODE=1", 1500); // Set Wi-Fi mode to Station
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("SSID:");
  lcd.print(ssid);
  lcd.setCursor(0,1);
  lcd.print("PWD:");
  lcd.print(password);
   sendCommand("AT+CWJAP=\"" + String(ssid) + "\",\"" + String(password) + "\"",
10000); // Connect to Wi-Fi
  //Serial.println("Connected to Wi-Fi!");
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("CONNECTED TO");
  lcd.setCursor(0,1);
  lcd.print("WI-FI");
}

void uploadToThingSpeak(int f1, int f2,int f3,int f4,int f5) {
      if      (sendCommand("AT+CIPSTART=\"TCP\",\"api.thingspeak.com\",80",
3000).indexOf("OK") != -1) {
    String httpData = "GET /update?api_key=" + apiKey + "&field1=" + String(f1) +
"&field2=" + String(f2) + "&field3=" + String(f3 ) + "&field4=" + String(f4 ) +
"&field5=" + String(f5 )+ " HTTP/1.1\r\n";
    httpData += "Host: api.thingspeak.com\r\n";
    httpData += "Connection: close\r\n\r\n";

    sendCommand("AT+CIPSEND=" + String(httpData.length()), 1000);
    sendCommand(httpData, 3000);
    //Serial.println("Data uploaded to ThingSpeak!");
  } else {
```

```
    //Serial.println("Failed to connect to ThingSpeak!");
  }
}

String sendCommand(String command, int timeout) {
  String response = "";
  Serial.println(command);
  long int time = millis();
  while ((time + timeout) > millis()) {
    while (Serial.available()) {
      char c = Serial.read();
      response += c;
    }
  }
 // Serial.println(response); // Debugging
  return response;
 }
```

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REAL-TIME ACCIDENT MONITORING SYSTEM USING IOT

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Abstract

This project involves the development of an IoT-based real-time accident monitoring system for enhanced accident analysis. The system integrates multiple sensors to continuously capture and log critical vehicular parameters, including speed, acceleration, geolocation, and environmental conditions. Upon detecting an accident, it autonomously activates an alert mechanism through a GSM module, transmitting notifications to emergency responders and predefined contacts. Simultaneously, it uploads detailed incident data to the cloud platform ThingSpeak, where it can be accessed for comprehensive analysis. The system's ability to store and transmit real-time data enables accurate accident reconstruction, identifying factors such as driver behavior, vehicle dynamics, and external conditions contributing to the incident. If the driver consumes alcohol, then the system stops the engine so that accidents due to alcohol consumption will be reduced. The collected data also facilitates post-accident investigations and helps improve safety measures. Additionally, the system supports remote diagnostics and real-time vehicle monitoring, providing valuable insights for enhancing road safety. This integrated approach aims to reduce emergency response times, improve the accuracy of accident investigations, and promote safer driving practices.

Keywords: Emergency Alerts, Drunken Driving Detection, GSM, Speed Monitoring, Acceleration, Over-Braking, Tilt Detection.

1. Introduction

1.1 Objective

The Internet of Things (IoT) has evolved into a transformative technology that revolutionizes the way we connect with and interact with the world around us. Through the connection of computer devices embedded in existing Internet infrastructure, IoT provides unprecedented connectivity, enabling seamless communication between devices, systems, and services. This connectivity goes far beyond traditional communication with machine-to-machine (M2M) interactions and includes a variety of protocols, domains, and applications. In virtually every automation area, IoT has paved the way for innovative solutions and advanced applications, unlocking new opportunities for efficiency, productivity, and convenience. From optimizing energy distribution to intelligent networks and transforming healthcare with cardiac monitoring implants, IoT has become an integral part of modern life. One particularly promising application of IoT technology lies in the field of vehicle security. The spread of intelligent objects and embedded sensors has enabled the development of sophisticated systems for accident detection and response. Smart Accident Detection and Alarm System is an advanced IoT-based solution designed to identify essential notifications for vehicle accidents and emergency services. The main goal is to improve survival rates by reducing emergency response times and ensuring prompt assistance. The system consists of multiple vehicle sensors that continuously monitor parameters such as vehicle speed, GPS coordinates, and accelerometer data. These sensor values are transmitted to a central server via wireless communication protocols, including 3G, 4G, Wi-Fi, or other available networks. The server employs machine learning algorithms to analyze incoming data and determine the occurrence of an accident. When an accident is detected, the system automatically sends an alert to emergency responders and predefined contacts, providing critical details such as location and accident severity. Additionally, notifications can be sent to victims' family members and friends via multiple communication channels, including SMS, emails, and push notifications. One major advantage of this system is its ability to function remotely, even in areas with limited or no cellular coverage. In such scenarios, satellite communications are used to ensure emergency alerts are transmitted successfully, enhancing the system's reliability in various environments. The system can also integrate with existing emergency services, enabling more efficient and coordinated responses. For instance, it can automatically notify nearby hospitals to prepare for incoming patients.

2. Literature survey

1. The GPS-GSM sheet, an inland navigation tracking system for automatic emergency recognition and location notification, by Shabrinet Althis, introduces a progressive vehicle tracking system (VTS) designed for domestic ships. Unlike traditional VTS used in cars, this system enables remote monitoring of ship movement and direction. It also includes real-time accident recognition features, utilizing continuous motion tracking to identify abnormal events. Once an irregularity is detected, the system immediately sends the ship's GPS coordinates to the relevant authority or owner, improving the rescue process and minimizing potential losses. The system relies on GSM communications for data transmission between onboard devices and the ground control center, limiting its use to inland waterways. However, replacing GSM with Iridium satellite communications would expand its usability to seaboard vessels, offering broader operational coverage. Additionally, this paper explores the concept of an integrated domestic waterway control system (IIWCS), incorporating this technology to enhance maritime security and navigation management.

2. Design and implementation of helmets to pursue accident zones, by Karthik P et al., focuses on improving road safety through an accident recognition system that automatically detects vehicle collisions and notifies pre-configured contacts such as family members, emergency services, police stations, and nearby hospitals. Increased urban complexity leads to worsening road conditions due to factors like traffic congestion, rising accident rates, and a high percentage of abandoned vehicles. These challenges contribute to higher transportation costs and inefficient vehicle movement. In densely populated countries such as India, delayed medical responses often result in fatalities due to a lack of timely assistance. This project aims to address these issues using GSM technology to send real-time vehicle location updates via SMS to designated recipients, ensuring immediate support. Additionally, the system offers vehicle tracking capabilities, making it particularly useful for fleet management. By regularly transmitting location data through GSM modems, businesses can efficiently monitor their vehicles, enhancing operational efficiency and security.

3. Jong-Ho Choi et al. propose a tracking methodology that integrates both traffic patterns and vehicle-specific characteristics, such as contextual background, localized positions, and vehicle measurements. A four-way overview tracking algorithm is used to determine the vehicle's initial position and size. To improve performance under varying lighting conditions, frame difference technology is employed to reduce the impact of brightness changes on contour detection. Each vehicle is represented using four key feature points, identified through the

minimum cover distance at the vehicle's location. By aligning the detected vehicle areas, the system effectively mitigates occlusion issues and enhances tracking performance.

3. Existing System

People in general or crisis management have the least foggy thoughts about the extent of closure and administration. Such data is lacking, which can cause some casualties. As a result, exploration tests arise in a way that answers how a crisis vehicle can be operated to reach a misfortune spot at the right time. Therefore, it is important to guide the vehicle and regulate movement signs according to the vehicle development. Many deaths have been occurring as the outbreak into the emergency vehicle mixture has changed. To achieve these obligations, we recommend running a system called the Entra Vectice Surveillance System (EMMS). Emergency vehicle checking systems can include GPS, GSM, and GIS management. GPS devices provide information about the location of the emergency vehicle and the shortest course. The main server that allows Google Manual to provide the shortest course between crisis vehicles and mixes. Therefore, we propose another outline to control activity signals and emergency vehicle reviews and obtain the above delegation so that the crisis vehicle has the ability to surpass all traffic control signals without maintaining itself. Each movement transition has a controller that controls the activity flow. Motion control is pointed out as a button, with each button having a GSM modem (a portable-enabled global system) connected to the controller. The hub is controlled by the main server by sending tax reports to a GSM modem (a global system for portable support). As the population grows, the number of vehicles on the road increases proportionally, increasing traffic congestion, leading to an increase in corresponding road accidents. The main challenge associated with such incidents is the delayed ambulance response when it reaches the scene of the accident or when the victim is transported to a medical facility. Timely medical interventions are extremely important for improving survival. To tackle this issue, it is important to implement automated accident recognition and warning systems that immediately notify rescue personnel and facilitate rapid transport of victims to hospitals. Additionally, timely accident reporting to investigators can accelerate the testing process and reduce delays in accident analysis and response measurements.

4. Proposed System

This work proposes an accident detection rescue system using IoT. IoT is very useful in this aspect as it replaces the conventional monitoring systems with a more efficient scheme. The proposed system of this project is shown in Fig 4.1.

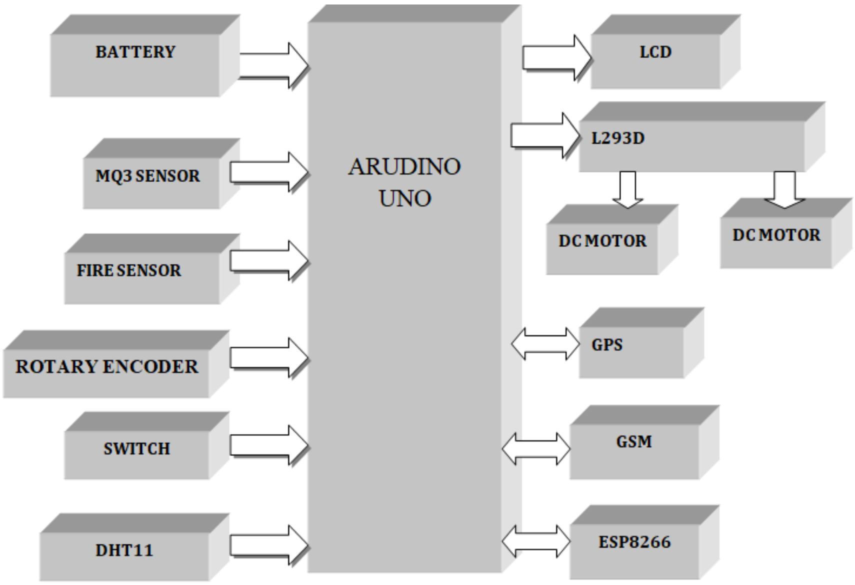

Fig 1 Block Diagram

Vehicle units include a microcontroller, MPU6050 movement sensor, MQ3 alcohol sensor, fire detection sensor, GPS module, and GSM communication module. The device is integrated into the vehicle to record accidents and transmit location data to designated recipients. The MPU6050 sensor continuously monitors sudden changes in vehicle movement, with the collected data processed by the microcontroller. The GPS module determines the real-time coordinates of the vehicle, identifies the accident location, and forwards this information to the GSM module. The GSM module then sends a warning message to pre-configured emergency contacts stored in the system, ensuring immediate notification in the event of an accident.

Flame sensors detect the presence of fire in the vehicle, while the gyro sensor monitors the alignment and angular speed of the vehicle. If any abnormal condition is detected, a warning message is displayed on the LCD (16*2), and the vehicle's location is sent to emergency contacts via the GPS and GSM modules. In critical situations, the braking system is activated immediately to ensure controlled deceleration, enhancing vehicle safety.

The Collision Anti-Crash Braking System (ACBS) is an advanced safety mechanism designed to detect potential collisions and autonomously activate the braking system. This feature slows down the vehicle or brings it to a complete stop, reducing the impact and risk of accidents. A switch is used to activate the system, ensuring the vehicle remains responsive without unnecessary sensitivity. The faster the wheel rotates, the higher the pulse frequency, indicating increased speed.

Additionally, the DHT11 sensor is used to measure the temperature inside the vehicle by reading the ambient temperature.

5. Hardware Components

5.1. ARDUINO UNO

The Arduino UNO is a widely used open-source microcontroller board, primarily developed with the Microchip Atmega328p processor by Arduino.cc. It features multiple digital and analog input/output (I/O) pins, allowing seamless integration with expansion modules and external circuits. The board includes 14 digital I/O pins and six analog input pins, providing a

Fig: Arduino UNO

versatile interface. Programming is done using the Arduino Integrated Development Environment (IDE) via a Type B USB connection

5.2 LCD (LIQUID CRISTAL DISPLAY)

A liquid crystal display (LCD) is a thin, flat display device composed of either colored or monochrome pixels placed in front of a reflective or light source. Each pixel consists of two polarizing filters with perpendicular polar axes and a layer of liquid crystal molecules between two transparent electrodes. When light passes through one filter, it is blocked by the other unless twisted by the liquid crystal, allowing controlled light transmission.

Fig: LCD (Liquid Crystal Display)

5.3 FIRE SENSOR

A flame sensor is a type of detector designed to identify and respond to fire or flames. It plays a critical role in fire detection systems, propane and natural gas connections, and alarm systems. Industrial boilers also use flame sensors to ensure proper operation. Compared to heat or smoke detectors, these sensors respond more quickly due to their flame-detection mechanism.

Fig Fire Sensor

5.4 ALCOHOL SENSOR:

The MQ-3 alcohol sensor is a specialized module designed to detect the presence of alcohol in the air. It utilizes a semiconductor material containing tin oxide, which changes its electrical conductivity when exposed to alcohol molecules. As alcohol comes into contact with the sensor, it alters the sensor's resistance, which is measured by its electronic circuit. MQ-3 sensors are commonly used in breathalyzers, automobile safety systems, and alcohol detection applications. They offer advantages such as high sensitivity, fast response time, and low power consumption.

5.5 GSM (GLOBAL SYSTEM FOR MOBILE COMMUNICATIONS)

The Global System for Mobile Communications (GSM) is a mobile network technology that enables mobile phones to connect to nearby cell towers. GSM networks operate on four different frequency bands. Short Messaging Service (SMS), commonly known as text messaging, was introduced as an inexpensive alternative to voice calls under GSM technology. A notable feature of GSM is the inclusion of a universal emergency number, 112, making it easier for foreign visitors to contact emergency services without knowing the local emergency number.

Fig: GSM Module

5.6 L2983

The L293 is a motor driver designed to provide bidirectional current flow of up to 1A with voltages ranging from 4.5V to 36V. The L293D variant supports bidirectional currents up to 600mA within the same voltage range. These motor drivers are commonly used to control solenoids, DC motors, and bipolar stepper motors, making them suitable for high-current and high-voltage applications.

5.7 DC MOTOR

The DC engine is designed to run on DC power. Two examples of pure DC design are Michael Faraday's homopole engine (unusual) and the engine containing the ball. This is (so far) novel. The most common DC engine types are brush and brushless types that use internal and external rectification to create alternating currents that vibrate from the DC source.

Fig: DC Motor

6. Results

The system is designed so that drunk driving can automatically be switched from the vehicle when the driver is deemed to be intoxicated. Additionally, in the event of a collision, the system will send the car's GPS location to the emergency contact. This means that rescuers can localize and assist passengers. When the vehicle fires, the system sends the vehicle's coordinates. This allows you to quickly determine the location of the fire and take appropriate measures. Overall, the system aims to improve traffic safety by preventing drunk driving and assisting in accidents and emergencies.

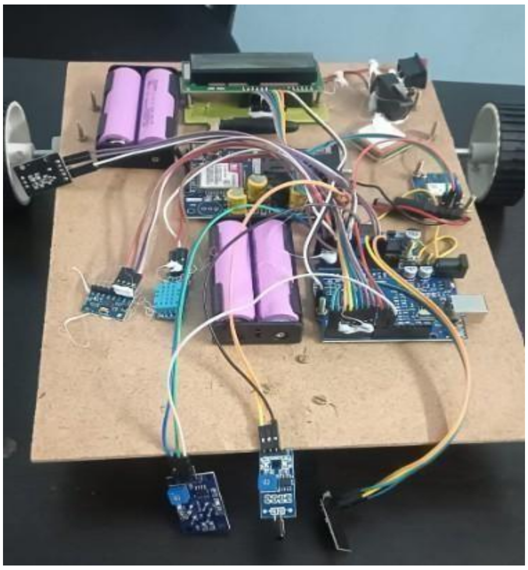

Fig: Hardware Setup

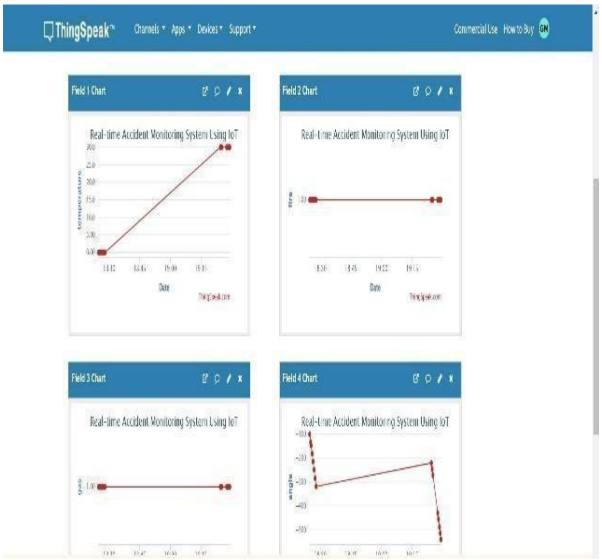

Fig: IoT platform output

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